Cheatsheet AZ-305 — 01 · Identity: Microsoft Entra ID licensing and governance

1. Licensing: what each tier unlocks

Feature	Free	P1	P2	Governance (add-on)
Users, groups, basic SSO, Entra Connect	✓	✓	✓	✓
Smart Lockout (brute force / password spray)	✓	✓	✓	✓
SSPR (password change)	✓	✓	✓	✓
Security Defaults (global MFA, legacy auth blocked, all-or-nothing)	✓	—	—	—
SSPR with **writeback** to AD DS	✗	✓	✓	✓
Conditional Access (conditions you define yourself)	✗	✓	✓	✓
Dynamic groups, hybrid join, on-prem Password Protection	✗	✓	✓	✓
Identity Protection (user risk / sign-in risk, risk-based CA)	✗	✗	✓	✓
PIM (eligible roles, JIT, approval, auditing)	✗	✗	✓	✓
Access Reviews (groups, apps, roles, guests)	✗	✗	✓	✓ (extended)
Entitlement Management (access packages)	✗	✗	✗	✓
Lifecycle Workflows (onboarding / offboarding)	✗	✗	✗	✓

Rule: "risk-based" → P2 · "just-in-time / eligible" → P2 · "access package" → Governance · "automatic onboarding" → Governance · "Free edition / minimize licensing" → look for something included by default (Smart Lockout, Security Defaults).

2. Identity Protection vs Identity Governance

	Identity Protection	Identity Governance
Question	Is this identity at risk right now?	Should this access exist?
Nature	Detection (ML) + response via CA	Lifecycle and recertification
Signals	user risk (leaked credentials), sign-in risk (impossible travel, anonymous IP/malware)	membership, roles, packages, joiners and leavers
Typical action	require MFA, block, force reset	activate temporarily, revoke, expire, deprovision
Frequency	continuous, every sign-in	periodic / on demand
License	P2	P2 (basic) / Governance (full)

3. The four pieces of Governance

Piece	What it does	Verb in the wording	Exam details
PIM	JIT activation of roles (Entra ID, Azure RBAC, groups)	"temporary", "just-in-time", "eligible", "no standing access", "justification"	Includes **alerts** (roles outside PIM, too many admins), **audit history** of activations and **access reviews of roles**. Recommended with MFA + approval
Access Reviews	Periodic recertification; whoever does not confirm loses access	"review periodically", "self-attestation", "remove if no response", "guests"	Reviewers: self, manager, designated. Configurable recurrence. Action on no response: remove
Entitlement Management	Access packages (groups + apps + sites) with request, approval and **expiration**	"access package", "request access", "partner / external collaboration", "expire"	Catalogs, policies per audience (internal / external), connected organizations
Lifecycle Workflows	Automates joiner / mover / leaver	"onboarding", "offboarding", "when a user leaves"	Triggers based on HR attributes (hire / leave date)

Access Reviews is part of Governance and embedded in PIM (roles) and Entitlement (packages). If the options only offer PIM for "review administrative roles + alerts + history" → PIM. If they offer Access Reviews for "groups / guests" → Access Reviews.

4. Conditional Access: anatomy of a policy

Block	What it contains	Examples
Assignments / Conditions	who, which app, from where, which device, which risk	users who manage production, app "Microsoft Azure Management", location, platform, sign-in risk
Grant	what is required to **grant** access	Require MFA · Require compliant device · Require hybrid joined device · Require approved app (AND / OR)
Session	what happens **after** granting	Sign-in frequency (e.g. 14 days) · Persistent browser session · App enforced restrictions

- "Require MFA / compliant / hybrid joined" → **Grant**.
- "Reduce MFA frequency following best practices" → **Sign-in frequency (Session)**, not the legacy "Remember MFA".
- "Register MFA methods" → **Authentication methods policy**, not CA.
- **Security Defaults** and custom CA are **mutually exclusive**.

5. Hybrid identity

	PHS	PTA	ADFS
Where the password is validated	Cloud (synced hash)	On-prem (agents)	On-prem (federation)
Survives on-prem network outage	**Yes**	No	No
Requirement "password hashes only on-prem"	✗	✓	✓
Enables Identity Protection leaked credentials	✓	Only if PHS as well	No
Recommendation	PHS (+ Seamless SSO). PTA with PHS as backup		

- **Entra Connect Health** → alerts on synchronization failures.
- **"License agreement failed"** when assigning a license → missing **Usage Location** (not synced from AD DS).
- **Custom domain:** must be public and verifiable with **TXT**; .local never works.
- **Non-routable UPN suffixes** → add a public UPN suffix before syncing.
- **Entra Domain Services** → LDAP / Kerberos / NTLM **inside Azure** without VPN to on-prem, fed from Entra ID.
- **Application Proxy + KCD** → publish an on-prem app with IWA/Kerberos, SSO, no VPN.
- **Lighthouse** → manage resources in **other tenants** without guest accounts. **B2B** → invite **users** as guests.
- **Workday provisioning:** 1 app can cover employees and contractors with conditional mapping (IIF); 1 provisioning agent per target domain (minimum).

6. Decision tree

```
What does the wording ask for?
├─ Detect risk / leaked credentials / impossible travel ..... Identity Protection (P2)
├─ Require MFA or device at sign-in ........................... Conditional Access · Grant (P1)
├─ How often to repeat MFA .................................... CA · Session · Sign-in frequency
├─ Which MFA methods can be registered ........................ Authentication methods policy
├─ Temporary role with justification / approval ................ PIM (P2)
├─ Periodically review groups / guests / apps .................. Access Reviews (P2)
├─ Review administrative roles + alerts + history .............. PIM (P2)
├─ Package of resources that is requested and expires ..... Entitlement Management (Governance)
├─ Automatic joiners and leavers ............................... Lifecycle Workflows (Governance)
├─ Brute force without a license ............................... Smart Lockout (Free)
└─ MFA for everyone without granularity and without a license ..... Security Defaults (Free)
```

7. Signals → answer

Wording	Answer
"risk-based", "leaked credentials", "atypical travel"	Identity Protection
"Just-In-Time", "eligible", "standing permissions"	PIM
"every three months", "self-attest", "removed if no response"	Access Reviews
"inactive admins list" (listing only)	Sign-in logs / Entra Workbooks
"access package", "request access", "expire after"	Entitlement Management
"hybrid joined device AND MFA"	Grant control in the existing policy
"reduce MFA prompts, 14 days, best practice"	Sign-in frequency
"Free tenant", "minimize licensing", "password spray"	Smart Lockout
"custom domain", ".local"	Public domain with TXT; .local is a distractor
"LDAP", "no VPN allowed"	Entra Domain Services
"MSP manages multiple tenants"	Lighthouse

8. Mistakes to avoid

1. Confusing P1 with P2: PIM, Identity Protection and Access Reviews are **P2**.
2. Choosing Identity Protection for "require MFA for everyone always" → that is CA Grant.
3. Choosing PIM for "periodic review of a group" when Access Reviews is among the options.
4. Choosing Session control to require MFA → Session governs after access is granted.
5. Choosing .local as a custom domain.
6. Choosing B2B when the wording forbids guest accounts.

Cheatsheet AZ-305 — 02 · Load balancers and Virtual WAN

1. The four load balancers

	Azure Load Balancer	Application Gateway	Azure Front Door	Traffic Manager
Scope	Regional	Regional	**Global**	**Global**
OSI layer	4 (TCP/UDP)	7 (HTTP/HTTPS)	7 (HTTP/HTTPS)	DNS (does not see traffic)
Protocol	Any TCP/UDP	HTTP(S) only, WebSocket	HTTP(S) only	**Any** (agnostic)
Sees the traffic	Yes (packets)	Yes (proxy)	Yes (proxy at the edge)	**No** (only resolves DNS)
SSL termination / offloading	✗	✓	✓ (at the edge)	✗
URL / host-based routing	✗	✓	✓	✗
WAF (OWASP)	✗	✓ (WAF v2)	✓	✗
mTLS (client cert)	✗	✓	✗	✗
Session affinity	Client IP	Cookie	Cookie	✗
Health probe	TCP / HTTP	HTTP(S) per path	HTTP(S) layer 7	Basic HTTP / TCP
Private IP / internal	✓ (Internal LB)	✓ (private frontend)	✗ (always public)	✗
Backends	VMs, VMSS, AKS nodes	VMs, VMSS, App Service, AKS (AGIC), IPs	App Service, AKS, VMs, Storage, other LBs	Any endpoint with IP/DNS
Typical case	SQL AG listener (ILB + Floating IP), L4 balancing	Regional web app with WAF and SSL	Global multi-region app with centralized SSL	Global DNS failover, non-HTTP protocols

Traffic Manager methods: Priority (active/standby) · Weighted · Performance (latency) · Geographic · MultiValue · Subnet. **Traffic Manager front-end failover** → **Endpoint monitor settings** (interval, timeout, tolerated failures).

2. How to choose in two questions

```
1. Is it HTTP(S)?
   ├─ No (SQL, RDP, FTP, generic TCP)
   │    ├─ Regional → Load Balancer
   │    └─ Global → Traffic Manager (the only global non-HTTP option)
   └─ Yes → 2.
2. One region or several?
   ├─ One region → Application Gateway (L7, WAF, SSL, mTLS, private IP)
   └─ Several regions → Front Door (L7, SSL at the edge, WAF, CDN)
                        Traffic Manager if DNS is enough and centralized SSL is not needed
```

Valid combinations: Front Door → App Gateway (regional) → VMs. Traffic Manager → App Gateway. Front Door → public Load Balancer.

3. Application Gateway: exam details

- **End-to-end SSL:** terminates TLS, **re-encrypts** towards the backend (backend HTTP setting with certificate) — vs **SSL offloading**: terminates and sends in clear text.
- **WAF v2:** OWASP CRS + custom rules incl. **rate limiting** (recent).
- **AGIC:** App Gateway as AKS ingress controller.
- Needs its **own (dedicated) subnet**.
- **v2** required for autoscaling, zone redundancy, private frontend.

4. Front Door: exam details

- SSL termination at global **points of presence**; managed certificates; custom domains.
- Automatic failover via **L7 health probes**; minimum latency via anycast.
- Integrated cache / CDN; WAF; rate limiting.
- Has **no** private IP; protects the backend with X-Azure-FDID / Private Link to origin (Premium).
- Decisive signal: **"SSL without configuring it on each backend"** + multi-region.

5. Load Balancer: exam details

- **Standard** SKU: zone-redundant, HA ports, NSG required, SLA 99.99 %. **Basic**: retired.
- **Internal LB for SQL AG listener:** enable **Floating IP** (DSR); **TCP** probe on the listener port; session persistence **does not apply**.
- **Outbound rules / NAT Gateway** for Internet egress.
- Pair of NVAs in HA → behind an **Internal LB** (avoids the UDR single point of failure).

6. Virtual WAN

Capability	Basic	Standard
Site-to-Site VPN	✓	✓
ExpressRoute	✗	✓
Point-to-Site VPN	✗	✓
Hub-to-hub (multi-region)	✗	✓
VNet ↔ VNet transit via hub	✗	✓
Secured virtual hub (Azure Firewall in the hub)	✗	✓
Third-party NVA in the hub	✗	✓
Upgrade Basic → Standard	Yes (mandatory first step before ExpressRoute)	—

- **Hubs:** one per region where presence is required ("supporting connection in three regions" → 3 hubs). Circuits and offices are spread across the existing hubs.
- **Global VPN profile (P2S):** the client connects **automatically to the nearest hub**.
- **Transitive routing:** native in Virtual WAN Standard; **classic peering is not transitive**.
- **FQDN filtering:** only **Azure Firewall** (secured hub) — NSGs and ASGs filter by IP/port.

7. Signals → answer

Wording	Answer
"regional", "path-based routing", "WAF", "SSL offloading"	Application Gateway
"multi-region" + "SSL managed centrally"	Front Door
"multi-region" + "any protocol" / "DNS"	Traffic Manager
"control the threshold for failing over the front end"	Traffic Manager endpoint monitor
"layer 4", "TCP/UDP", "internal", "SQL listener"	Load Balancer (+ Floating IP)
"single private IP + mTLS + rate limiting" (APIM options)	APIM Premium Internal (exam answer)
"ExpressRoute" + "Virtual WAN"	Standard
"P2S closest region", "transitive", "FQDN filtering"	Virtual WAN Standard + secured hub
"connection in N regions" → hubs	N hubs

8. Mistakes to avoid

1. Choosing Traffic Manager when **centralized SSL** is required → it does not see the traffic.
2. Choosing Load Balancer to route by URL → it is layer 4.
3. Choosing Front Door for **private** access → it is always public.
4. HTTP probe on the SQL listener → TCP.
5. Session persistence on the ILB of an AG listener → what is missing is Floating IP.
6. Virtual WAN Basic with ExpressRoute → impossible; Standard.
7. Counting circuits or offices as hubs → count the **required regions**.

Cheatsheet AZ-305 — 03 · Azure Storage complete

1. Redundancy: copies and what it protects against

Type	Copies	Where	Protects against	Read from secondary
LRS	3	1 datacenter	Disk / node failure	—
ZRS	3	3 zones (same region)	Failure of **one datacenter / zone**	—
GRS	6	3 LRS primary + 3 LRS paired region (async)	**Region** failure	✗
RA-GRS	6	same as GRS	Region failure	✓ -secondary endpoint
GZRS	6	3 ZRS primary + 3 LRS secondary	Zone **and** region	✗
RA-GZRS	6	same as GZRS	Zone and region	✓

- "datacenter failure" → **ZRS** is enough. "region failure" / "maximize resiliency" → **GRS / GZRS**.
- "replicates" ≠ "read from replica": if reading from the secondary is not required, **GRS** (cheaper than RA-GRS).
- **Premium (SSD) only supports LRS and ZRS.** Never GRS.

2. Account failover cycle (customer-managed)

```
Before: configure GRS/RA-GRS (no secondary, no failover)
   → Initiate failover (the secondary becomes primary; the endpoint does not change for apps)
   → The account is automatically left in LRS
   → Reconfigure GRS to be able to fail back
   → Failback once re-replication completes
```

Exam sequence: **4 (GRS before) → 2 (failover) → 1 (GRS after)**. ZRS never appears.

3. Account types

Type	Performance	Services	Hot/Cool/Archive tiers	Redundancy	When
General-purpose v2 (StorageV2)	Standard (HDD)	Blob, Files, Queue, Table, ADLS Gen2	✓	All	**Default** for everything new
General-purpose v1	Standard	Blob, Files, Queue, Table	✗	Limited	Legacy; distractor
Blob Storage (legacy)	Standard	Blobs only	✓	All	Legacy; StorageV2 replaces it
BlockBlobStorage (Premium block blobs)	**Premium SSD**	Block / append blobs	✗ (single tier)	**LRS / ZRS**	Maximum transactions and minimum latency on blobs
Premium page blobs	Premium	Page blobs	✗	LRS / ZRS	Unmanaged VM disks
Premium file shares (FileStorage)	Premium	Azure Files (SMB/NFS)	✗	LRS / ZRS	High-performance file shares

- **GPv1 + Premium** does not exist (invalid option).
- **Data Lake Storage Gen2** = StorageV2 (or Premium block blob) with **hierarchical namespace**: ABFS/Hadoop driver, **POSIX ACLs** per folder and file, real directories.
- "Hadoop", "HDFS", "POSIX ACL" → GPv2 with hierarchical namespace.

4. Blob access tiers

Tier	Storage cost	Access cost	Read latency	Minimum retention	Use
Hot	High	Low	ms	—	Frequent access, active uploads/downloads
Cool	Medium	Medium	ms	30 days	Infrequent access, immediate access
Cold	Low	High	ms	90 days	Very infrequent, immediate access
Archive	Minimum	Maximum	**Hours (rehydration, up to 15 h)**	180 days	Almost never; can wait

- "available within 24 hours" / "rarely accessed" + "minimize cost" → **Archive** (blobs only, GPv2 / Blob Storage; not Files, not GPv1).
- "optimized for uploads/downloads" + "lowest cost" → **Hot** on Standard (the saving comes from Standard vs Premium, not from Cool).
- The account's **default tier** does not matter if each blob is explicitly set to Archive.
- **Lifecycle management** moves blobs between tiers and deletes them by age rules.

5. Data protection on blobs

Feature	What it does	Time direction	Prerequisites
Soft delete (blob / container)	Keeps deleted items for N days	Recover what was deleted	—
Versioning	Saves every version on overwrite	Recover previous versions	—
Change feed	Ordered, immutable log of all changes	Know what changed and when	—

Feature	What it does	Time direction	Prerequisites
Point-in-time restore	Return a container to a point in time	Go back	**Versioning + soft delete + change feed**
Immutable storage (WORM)	Prevents modifying / deleting during retention	Prevent changes going forward	Versioning (at version level)
Object replication	Async copy of blobs to another account	Replicate going forward	Versioning + change feed
Resource lock (CanNotDelete)	Prevents deleting the resource (account), not the blobs	ARM resource protection	—

- **Immutable: time-based retention** (X years; **Locked** for regulatory, Unlocked can be shortened) and **legal hold** (until removed). "new data can be written, modification prevented for 3 years" → time-based retention.
- PITR and immutability are incompatible on the same container.

6. Access and security

Mechanism	What it is	When
Account keys	Two master keys	Avoid; rotate; never in clients
SAS (account / service / user delegation)	Token with permissions, expiry, IP	Temporary access for clients; **stored access policy** = centralized revocation
Entra ID RBAC (Storage Blob Data Reader/Contributor)	Roles on the data plane	Preferred for apps with managed identity
Private Endpoint	Private IP in the VNet + Private DNS zone	"no public endpoint", access from on-prem via ExpressRoute/VPN
Service endpoint	The subnet reaches the service over the Microsoft network; the service keeps a public IP	When a private IP is not required
Account firewall	Allowed IPs / VNets	Complement
SSE	Encryption at rest always on; **PMK** by default or **CMK** in Key Vault	"manage own keys" → CMK
Secure transfer required	Forces HTTPS	Encryption in transit

7. Azure Files and File Sync

- **Azure Files:** SMB / NFS; tiers Premium, Transaction Optimized, Hot, Cool (**no Archive**). Identity: Entra Domain Services, on-prem AD DS or Entra Kerberos.
- **Azure File Sync:** local cache on Windows Server + cloud tiering; continued file access if a site goes down.
- **Azure File Sync ≠ backup**: to copy, use Azure Backup for file shares.

8. Bulk transfer

Tool	Mechanism	When
AzCopy	Copy over the network	Moderate volumes, scripting
Import/Export	You ship **your own disks** encrypted with BitLocker	TB with little bandwidth, own hardware
Data Box (Disk / Data Box / Heavy)	**Microsoft ships a device**	Tens of TB to PB
Data Box Gateway	Virtual appliance that sends to Azure continuously	Continuous transfer from on-prem
Data Share	Share snapshots with **another organization** in a governed way	External partner, another tenant

9. Decision tree (account + redundancy + tier)

```
Maximum performance / minimum latency on blobs?
├─ Yes → BlockBlobStorage Premium → ZRS (datacenter) or LRS redundancy
└─ No → StorageV2 Standard
        ├─ Maximum resiliency → GRS / RA-GRS (RA only if reading is required)
        ├─ Datacenter only → ZRS
        ├─ Hadoop / POSIX → hierarchical namespace (ADLS Gen2)
        └─ Tier: frequent → Hot · infrequent immediate → Cool · can wait hours → Archive
```

10. Signals → answer

Wording	Answer
"highest transaction rate, lowest latency"	BlockBlobStorage Premium + ZRS
"lowest cost per GB" + "uploads/downloads"	GPv2 Standard Hot
"maximize data resiliency"	StorageV2 + RA-GRS (never Premium)
"available within 24 hours"	Archive
"prevent modification for N years"	Immutable, time-based retention (Locked)
"point-in-time restore" (versioning and soft delete already enabled)	Change feed
"fail over and fail back"	GRS before → failover → GRS after

Wording	Answer
"Hadoop-compatible", "POSIX ACLs"	GPv2 + hierarchical namespace
"REST" + "20 independent tables" + "second region" + "min cost"	Table Storage + GRS
"public endpoint must be prevented"	Private Endpoint
"manage own encryption keys"	SSE with CMK (Key Vault)
"ship physical disks", "own hardware"	Import/Export
"share snapshots with partner company"	Data Share

11. Mistakes to avoid

1. Premium + GRS → does not exist.
2. RA-GRS when nobody asks to read from the secondary → GRS is cheaper.
3. Archive on Azure Files or on GPv1 → does not exist.
4. Cool for "optimized for uploads/downloads" → penalizes transactions; Hot.
5. Forgetting that failover leaves the account in LRS.
6. Confusing Data Box (Microsoft's device) with Import/Export (your own disks).

Cheatsheet AZ-305 — 04 · Governance: Policy, Blueprints, RBAC, management groups and tenants

1. Hierarchy and the tenant wall

```
Microsoft Entra ID tenant  ← absolute boundary: nothing crosses it
└── Root management group
    └── Management groups (up to 6 levels)
        └── Subscriptions
            └── Resource groups
                └── Resources
```

- **RBAC, Policy, Blueprints and management groups inherit downwards** and **never cross tenants**.
- "Minimum number of assignments / management groups / definitions" → **count tenants** and apply at the root MG of each one.
- Example: 2 tenants with 10 + 5 subscriptions → Network Contributor with **2 assignments** (one per root MG).
- **Resource groups** do not limit the technical association between resources (an NSG in RG2 can be associated with a subnet in RG1); they are only lifecycle, RBAC and Policy scope. The subscription and the region do limit it.
- **Tags**, not resource groups, to group by business dimension (department, cost center). A resource can only be in one RG.

2. Azure Policy: effects

Effect	What it does	New resources	Existing resources	Remediation task	Identity required
Deny	Blocks creation / update	✓ blocks	Flags as non-compliant	✗	—
Audit	Only logs non-compliance	✓ logs	✓ logs	✗	—
AuditIfNotExists	Audits if a related resource is missing	✓	✓	✗	—
Append	Adds fields / tags at creation; **denies on conflict**	✓	✗	✗	—
Modify	Adds / replaces / removes **tags and properties** of the resource	✓	✓	✓	Managed identity + **Contributor**
DeployIfNotExists (DINE)	Deploys an **ARM template** if something is missing (sub-resource, extension, diagnostic setting)	✓	✓	✓	Managed identity + deployment role
Disabled	Disables without deleting	—	—	—	—
EnforceOPAConstraint / EnforceRegoPolicy	Only the **Kubernetes/AKS** add-on (Gatekeeper/OPA, Rego)	—	—	—	—

- Tags inherited from the RG, new and existing, with remediation → **Modify**.
- Enable TDE with a template, AMA agent, diagnostic settings → **DeployIfNotExists**.
- Only at creation and without remediation → Append.
- Remediation: **managed identity** (not service principal) with the **minimum role** (Contributor for tags; never User Access Administrator, which manages RBAC).
- Evaluation: on assignment, when a resource changes and **every 24 h**; on demand only via **CLI (az policy state trigger-scan) / PowerShell / REST**, not from the portal.
- Non-compliance alerts → diagnostic setting to Log Analytics + **Log** type alert.
- Policy ≠ RBAC: Policy governs configurations; RBAC governs permissions. Policy cannot assign roles.
- Policy at management group level with **exclusions** for specific subscriptions.
- An **Initiative** groups definitions so they can be assigned together.

3. Azure Blueprints (being retired, replaced by Template Specs / Deployment Stacks)

Element	Where it is created	How many	Counted by
Management group	One per tenant (minimum)	= tenants	Tenants
Blueprint definition	In the management group (for reuse) or in a subscription	= hierarchies where it is reused	Tenants
Blueprint assignment	**Per target subscription**	= target subscriptions	Subscriptions

- Artifacts: resource groups, ARM templates, **role assignments**, policy assignments.
- Example East (Sub1, Sub2) / West (Sub3, Sub4): **2 MG · 2 definitions · 4 assignments**.
- "Minimize definitions" → define at the **root MG** of each tenant. "Assignments" → one per subscription.
- For RBAC + Policy + repeatable structure at scale → Blueprints (or Bicep + Deployment Stacks).

4. RBAC

Concept	Exam detail
Owner	Everything + manage assignments. Required to **change the directory of a subscription**
Contributor	Everything except RBAC. Enough to remediate tags, create a Recovery Services vault
Reader	Read only
User Access Administrator	Only manages RBAC
Network Contributor	Networking; apply at the root MG of each tenant
Custom roles	Define at the **management group** to reuse across all subscriptions; DataActions for the data plane (blobs, files)
Global Administrator + "Access management for Azure resources"	**Elevate access**: User Access Administrator at / ; for subscriptions with no Owner. Via PIM and revert afterwards
Site Recovery Contributor / Operator	Operate replication / run failover; neither creates the vault (that is Contributor)
Resource-context vs workspace-context (Log Analytics)	Resource-context: see only logs of the resources where you have RBAC; recommended for hub-spoke
Resource locks	CanNotDelete / ReadOnly; apply even to Owners; inherited
Resource Guard	In **another subscription/tenant**; requires a role there for critical backup operations (anti-ransomware, separation of duties)

5. Cost Management and organization

- **Tags** + Cost Management → reports per department; Policy (Modify) guarantees they exist.
- **Budgets** with alerts; **Azure Advisor** recommendations (does not act).
- **Management groups** to structure EA / divisions; **subscriptions** as billing and quota boundary.

6. Decision tree

```
What is asked?
├─ Block a configuration ..................................... Policy Deny
├─ Only detect ............................................... Policy Audit
├─ Add/fix tags or properties, new and existing ............... Policy Modify + managed identity + Contributor
├─ Deploy something missing with an ARM template ............. Policy DeployIfNotExists
├─ Governance in an AKS cluster .............................. Policy add-on (OPA/Rego)
├─ Permissions for all subscriptions with minimum effort ..... RBAC at root MG (one per tenant)
├─ Reuse a custom role across many subscriptions .............. Define at MG
├─ Repeatable package RBAC + Policy + structure .............. Blueprints (def. at MG, assignment per sub)
├─ Prevent deleting an RG even for Owners ..................... Resource lock / Policy Deny delete
├─ Subscription with no administrators after moving tenant ..... Global Admin + elevate access
├─ Move a subscription to another tenant ...................... Owner on the subscription
├─ Protect backups from compromised administrators ............. Resource Guard in another subscription
└─ Group resources by department for reporting ................ Tags (+ Policy)
```

7. Signals → answer

Wording	Answer
"minimum number of role assignments" + management groups	Number of tenants
"minimum management groups / blueprint definitions"	Number of tenants
"minimum blueprint assignments"	Number of target subscriptions
"remediation tasks" + "tags"	Modify, managed identity, Contributor
"remediation tasks" + "ARM template"	DeployIfNotExists
"trigger on-demand compliance scan"	Azure CLI / PowerShell / REST
"change the directory of a subscription"	Owner
"no accounts have permissions on the subscription"	Global Administrator (elevate access)
"create and manage the Recovery Services vault"	Contributor
"group resources by department" + "compliance reports"	Tags + Azure Policy
"ransomware" + "require a role in another subscription"	Resource Guard
"MSP manages other tenants"	Lighthouse

8. Mistakes to avoid

1. Assigning Blueprints at the management group → **assignments are per subscription**; the **definition** goes at the MG.
2. Append for existing resources → does not remediate; Modify.
3. Service principal for remediation → managed identity.
4. User Access Administrator to write tags → Contributor.
5. Believing that an MG or a role crosses tenants.

6. Reorganizing resource groups to group by department → tags.
7. Looking for the "evaluate now" button in the portal → it does not exist.

Cheatsheet AZ-305 — 05 · OAuth, application identities, Key Vault and API Management

1. Roles in OAuth 2.0 with Microsoft Entra ID

Role	Who	What it does
Authorization Server	Microsoft Entra ID (or B2C)	**Authenticates, asks for consent and issues/signs the tokens**
Client	The calling app (web app, SPA, service)	Requests the token and sends it in Authorization: Bearer
Resource Server	The API	**Validates** the token (signature, aud , expiry, claims) and decides **authorization**
Resource Owner	The user	Gives consent

- "Access tokens generated by" → **Entra ID**. "Authorization decisions performed by" → **the Web API**.
- Key Vault never issues OAuth tokens.

2. Grant types

Grant	User?	Client secret	Use	Status
Authorization Code (+PKCE)	Yes	Yes (PKCE if it cannot store it)	Web apps, mobile, SPAs	✓ Recommended
Client Credentials	**No**	Yes (secret / certificate)	App-to-app, jobs, backends, managed identities	✓ For M2M
Implicit	Yes	No	(Legacy) SPAs	✗ Deprecated
Resource Owner Password	Yes (the app sees the password)	Yes	(Legacy) migrations	✗ Avoid; incompatible with MFA
Device Code	Yes	No	Devices without a browser	Niche

- **state parameter:** opaque value that goes out and comes back in the redirect; CSRF protection and **transport of the app's own data ("custom data")**. Only makes sense with a redirect (Auth Code / Implicit).

3. Delegated vs Application permissions

	Delegated	Application
Acts on behalf of	**A signed-in user**	**The app** (no user)
Flow	Authorization Code	Client Credentials / Managed Identity
Consent	User or admin	**Always admin consent**
Claim in the token	scp (scopes)	roles (app roles)
Effective scope	What the user can do	Everything granted, across the whole organization
Defined in the API as	Scope (Expose an API)	App role (App roles, member type Applications)

4. App registration: which blade each app touches

Need	App	Blade
Expose permissions/roles to other apps	The **API**	**Expose an API** (scopes) / **App roles**
Request permissions/roles of another API (so they appear in the token)	The **client**	**API permissions** (+ admin consent)
Add optional claims to the token	Any	Token configuration
Register an app for Entra ID authentication	The app	App registration (the basis of everything)
Publish an on-prem app with SSO without VPN	—	Application Proxy + Enterprise Application (KCD for Kerberos)
App registration vs Enterprise application	Registration = definition (one); Enterprise app = instance/service principal in the tenant (allows assigning users, CA, SAML SSO)	

- Silent SSO on **Entra-joined** devices → app registration + the device's **PRT**; nothing else is needed.
- Access only from company computers → Conditional Access · Grant · compliant / joined device.

5. Identities for applications

Identity	Where the app runs	Credentials	When
Managed identity (system / user-assigned)	**Azure resources only** (VM, App Service, Functions, AKS, Policy remediation, Data Factory)	None to manage	Whenever the app is in Azure
Service principal	Anywhere (on-prem, another cloud, CI/CD)	Secret or certificate	App outside Azure; staff rotation
Personal access token / user	—	Tied to a person	Avoid (staff turnover)

Identity	Where the app runs	Credentials	When
Certificate in Key Vault	Apps that need mTLS or signing	Managed certificate	App authentication with cert

- **IMDS** (169.254.169.254/metadata/identity/oauth2/token): local endpoint through which a **VM** obtains the token of its managed identity. "Which endpoint to obtain an access token" with managed identity → IMDS.
- **User-assigned** MI: reusable across several resources, independent lifetime. **System-assigned**: created and deleted with the resource.

6. Key Vault: three objects

Object	What it is	Operation	Examples
Secret	Sensitive text retrieved as is	GET secret	Connection strings, passwords, API keys
Key	Cryptographic key (RSA/EC) that **never leaves**	encrypt / decrypt / sign / wrap	CMK for Storage / SQL TDE / ADE, signing
Certificate	X.509 certificate with its key and renewal	GET certificate , auto-renew	TLS, mTLS, app signing

- **Standard SKU** (software) vs **Premium** (**HSM-backed keys, FIPS 140-2 Level 2**) vs **Managed HSM** (Level 3, single-tenant).
- Access: **RBAC** (recommended) or access policies; the app's **managed identity** as caller; **Private Endpoint** for private access.
- "Protect connection strings, only the app at runtime, no hard-coded credentials" → **Key Vault** (the store), with managed identity as the access mechanism.

7. Encryption: where each technology sits

State	Technology
At rest	SSE (Storage, PMK/CMK), **TDE** (SQL, files), **Azure Disk Encryption** (BitLocker/DM-Crypt + customer Key Vault, auditable), Always Encrypted (columns; only the client decrypts)
In transit	TLS/HTTPS, IPsec (VPN), MACsec (ExpressRoute Direct)
In use	**Azure Confidential Computing** (TEE, Confidential VMs)
Masking by role	**Dynamic Data Masking** (some see everything, others the last 4 digits)

- mTLS in App Service: the client certificate arrives in the **X-ARR-ClientCert** header.
- P2S VPN with certificates: upload the **root public key** to Azure; client certificates stay on the devices; individual revocation by thumbprint.

8. API Management

Concept	What it is
Products	Packaging of APIs for consumers: subscriptions, quotas, terms
Policies	XML in inbound / backend / outbound / on-error: validate-jwt , rate-limit , quota , set-header , ip-filter , cors , validate-client-certificate , XML↔JSON transformation
VNet modes	**External**: gateway with public IP + private backends · **Internal**: everything private · **None**: no VNet
Tiers	Consumption (serverless) · Developer (not for production) · Basic · Standard · **Premium** (VNet inject, multi-region, ISE-like) · v2 tiers (Basic v2, Standard v2 with outbound VNet integration)
VNet injection (External/Internal)	Only **Developer and Premium** (classic)
Self-hosted gateway	APIM gateway deployed on-prem / other clouds

- "third-party OAuth provider" + different rate limits per consumer + "no changes to backend" + "no guest accounts" → **APIM** (validate-jwt + Products).
- "remove AspNet-Version header" → **policy** (set-header delete).
- "private IP + mTLS + rate limiting" (APIM options) → **Premium Internal** (exam answer; Standard does not inject into a VNet).
- Chain Front Door → APIM → backend: **OWASP → Front Door; JWT/ratelimit/products → APIM; IP filtering per-API → APIM** (in matching, B2C gets "JWT").
- Functions HTTP trigger: **Anonymous** (public) · **Function** (key for trusted apps) · **Admin** (master key). Read-only → **GET only**; CORS is handled by the platform (do not add OPTIONS).

9. Decision tree (app identity)

```
Where does the app run?
├─ In Azure ............................ Managed identity (token via IMDS on VMs)
├─ Outside Azure ....................... Service principal (secret/certificate in Key Vault)
└─ Does it act on behalf of a user?
     ├─ Yes → Authorization Code, delegated permissions (scp)
     └─ No → Client Credentials, application permissions (roles), admin consent
```

10. Signals → answer

Wording	Answer
"tokens generated by"	Microsoft Entra ID
"authorization decisions by"	The Web API
"custom data in the grant flow"	Support state parameter
"token must include the Writer role of App1" (modify App2)	API permissions of App2
"grant permissions using Entra-generated claims"	Microsoft Entra ID
"JWT validation policy using"	Azure API Management
"no user interaction", "service to service"	Client Credentials / application permission
"obtain access token" + managed identity on a VM	IMDS
"on-premises app" + "staff turnover"	Service principal
"retrieve keys for use in code" (Development team)	Managed identity
"connection strings", "runtime only"	Key Vault (secrets)
"FIPS 140-2 Level 2 HSM"	Key Vault Premium
"verify client certificate in web app"	HTTP request header
"data in use"	Confidential Computing
"supervisors see full number, operators last 4 digits"	Dynamic Data Masking
"never in plain text in the DB, only client decrypts"	Always Encrypted

11. Mistakes to avoid

1. Managed identity for an on-prem app → it does not exist outside Azure.
2. Managed identity as a secret "store" → it is the access mechanism; the store is Key Vault.
3. Confusing app registration (definition) with enterprise application (instance).
4. Implicit or ROPC as the "recommended" flow.
5. Modifying App roles in the client app → API permissions is what gets modified.
6. APIM Standard for a private IP → it does not inject into a VNet.

Cheatsheet AZ-305 — 06 · Messaging and events

1. The four services

	Queue Storage	Service Bus	Event Hubs	Event Grid
Type	Simple queue	Enterprise broker (queues + topics)	Event streaming	Push notifications
What it carries	Small messages (64 KB)	Messages (256 KB / 100 MB Premium)	Massive data/telemetry streams	Discrete events ("something happened")
Model	Pull	Pull (queues) / pub-sub (topics)	Pull, partitioned, consumer groups	**Push** to subscribers
Volume	Very high (up to 500 TB)	High	**Millions/sec**	Individual events
Guaranteed order (FIFO)	✗ (best effort)	✓ with **sessions**	Per partition	✗
Transactions / duplicates / dead-letter	✗	✓ (dedup, DLQ, transactions)	✗	24 h retries, optional dead-letter
Retention	Up to 7 days	Until configurable expiry	1–7 days (more in Premium/Dedicated), **replayable**	24 h retries
Protocols	REST	AMQP, REST	AMQP, Kafka, REST	HTTPS, webhooks, CloudEvents
Typical producers	Apps	Apps	Apps, devices, **diagnostic settings**	Azure resources (Blob, ARM, Key Vault, IoT Hub), custom topics
Typical consumers	Apps, Functions	Apps, Functions, Logic Apps	Stream Analytics, Functions, Databricks, Synapse	Functions, Logic Apps, webhooks, Event Hubs
Typical case	Cheap queue between microservices	Orders, FIFO, pub/sub, integration	IoT telemetry, logs, clickstream	"When a blob is created, run X"

2. Service Bus: exam features

Feature	What it does	Signal
Sessions	Groups messages by SessionId; **guaranteed FIFO** and one consumer per session	"FIFO", "process in order"
Partitioning	Spreads the queue across brokers → more throughput; **breaks global order**	"throughput", never "FIFO"
Topics + subscriptions	Pub/sub with filters per subscription	"multiple subscribers", "filter"
Auto-forwarding	Chains topic/subscription → destination queue	"forward to individual queues"
Dead-letter queue	Messages that cannot be processed	"poison messages"
Duplicate detection	Discards duplicates by MessageId	"exactly once"
Scheduled delivery / deferral	Deliver later / postpone	"delay"
Premium	Dedicated resources, VNet/Private Endpoint, large messages	"isolation", "large messages"

3. Event Grid vs Event Hubs (the usual confusion)

Question	Event Grid	Event Hubs
Who produces?	Azure resources that support it natively, or your app (custom topic)	Apps, devices, **diagnostic settings of any service (incl. Entra ID)**
What do I want to do?	**React** to each event	**Ingest, retain and analyze** a stream
Size / frequency	Small, sporadic	Large, continuous
Mental phrase	"Tell me when something happens"	"Give me everything that happens, raw"

- **Entra ID does not publish to Event Grid.** Its logs only come out via **diagnostic settings** → Log Analytics / Storage / **Event Hubs**.
- Pipeline "Entra ID → ? → ? → Cosmos DB": **Event Hubs → Functions → Cosmos DB**.
- **Event Grid runs C# code with access to a private IP + minimum cost → Functions Premium** (Consumption has no VNet integration; Logic Apps ISE is very expensive).

4. Diagnostic settings: the three destinations

Destination	What for	Signal
Log Analytics	Query (KQL) and **alert**	"analyze", "alerts"
Storage account	Cheap long-term archiving	"retain for 7 years", "archive"
Event Hubs	**Streaming** to SIEM / external system / pipeline	"third-party SIEM", "stream", "Cosmos DB"

5. Other integration services

Service	What it is	Signal
Logic Apps	Low-code orchestration with connectors; Consumption vs Standard (VNet) vs ISE (isolated, expensive)	"no code", "connectors", "workflow"
Functions	Code in response to events; Consumption / Premium (VNet, no cold start) / Dedicated	"custom code", "C#"
Durable Functions	Stateful workflows; **Orchestrator** defines the flow and the **durable timers**; Activity does the work; Client starts it	"durable timer" → Orchestrator
On-premises Data Gateway	Logic Apps / Power BI / Power Automate access on-prem SQL without opening ports	"no inbound ports"
Self-hosted Integration Runtime (formerly Data Management Gateway)	Data Factory / Synapse access on-prem data	"ADF + on-premises"
Notification Hubs	Push to mobile (iOS/Android)	"mobile push"
SignalR Service	Real time to browsers	"real-time web"
Azure Cache for Redis	In-memory cache, distributed **session state** (serializable), close to the app	"session state", "serializable", "near the app"
CDN	Static content close to the **user**	"near the user"

6. Decision tree

```
What do I need to move?
├─ A notice that something happened in Azure, to react ....... Event Grid
├─ A massive data stream to analyze / retain ................. Event Hubs
├─ Entra ID logs / diagnostic settings to the outside ........ Event Hubs
├─ Messages with FIFO / transactions / pub-sub / DLQ ......... Service Bus (sessions for FIFO)
├─ Simple, cheap, REST queue, no guarantees .................. Queue Storage
├─ Push to mobile ............................................ Notification Hubs
└─ Scalable web session state ............................... Azure Cache for Redis
```

7. Signals → answer

Wording	Answer
"FIFO"	Service Bus + sessions
"forward to individual queues from a topic"	Auto-forwarding
"simple REST queue between microservices"	Queue Storage
"user creation and role assignment events → Cosmos DB", service 1	Event Hubs (service 2: Functions)
"blob created → process it"	Event Grid
"IoT telemetry, millions of events"	Event Hubs
"code in response to Event Grid + private IP + min cost"	Functions Premium
"durable timer"	Orchestrator function
"Logic App → SQL on-prem, no inbound ports"	On-premises Data Gateway
"session state, serializable, scalable"	Azure Cache for Redis (clustered)

8. Mistakes to avoid

1. Event Grid for Entra ID logs → it is not a diagnostic settings destination.
2. Partitioning for FIFO → breaks the order; sessions.
3. Queue Storage when order or transactions are required.
4. Functions Consumption to reach a private IP.
5. Output Cache provider confused with Session State provider (Redis).

Cheatsheet AZ-305 — 07 · Monitoring

1. Who does what

Service / feature	What it is	Signal in the wording
Azure Monitor	Umbrella platform: metrics, logs, alerts, workbooks	generic
Log Analytics workspace	Log repository queryable with **KQL**; correlates different sources; basis for Log-type alerts	"correlate resource data with application data", "query logs", "alerts from logs"
Application Insights	APM: requests, dependencies, exceptions, **availability tests** (synthetic, from the Internet), Application Map, response-time alerts	"simulate user connections/transactions", "response time goals", "performance of the app"
Activity Log	**Control plane** audit (who created/modified/deleted/stopped what)	"who stopped the VM", "resource changes"
Service Map / VM Insights	Topology and dependencies between processes and servers	"dependencies between servers"
Network Watcher	Network diagnostics: IP flow verify, NSG diagnostics, next hop, connection troubleshoot, packet capture, NSG flow logs	"packets allowed or denied", "next hop", "capture traffic"
Workbooks	Interactive reports over logs (e.g. "Inactive user accounts")	"list inactive admins"
Azure Advisor	Recommendations (cost, security, reliability, performance); **does not act**	"recommendations"
Defender for Cloud	Security posture, JIT VM access, threat detection	"secure score", "JIT ports"
Connection Monitor	Latency/availability between endpoints (incl. ExpressRoute)	"connectivity over time"

2. Agents and collection

Element	What it is
Azure Monitor Agent (AMA)	Current agent for Azure VMs and (via **Azure Arc**) on-prem/other clouds. Replaces MMA/OMS
Data Collection Rule (DCR)	What to collect (Windows events, IIS, perf, syslog, custom), from which machines (associations, **any region**), where to (workspace)
Data Collection Endpoint (DCE)	Regional endpoint; **required only with Private Link** or custom/text logs
Azure Arc	Brings external servers in as Azure resources → AMA, Policy, Update Manager
Diagnostic settings	PaaS resources and Entra ID → Log Analytics / Storage / Event Hubs

- **Minimum DCRs:** count **distinct sets of data types**, not regions. Windows logs (Central+East) and IIS logs (East+West) → **2 DCRs**; East has both associated. Workspace over public endpoint = no regional DCE.
- **Centralized** workspace for hybrid monitoring; **resource-context RBAC** so each team sees only its own resources.

3. Private Link for Azure Monitor (AMPLS)

```
Workspace1 ── 1 AMPLS ─┬── Private Endpoint (VNet1 or VNet2, peered → serves both)
                       └── Private Endpoint (VNet3, isolated)
```

- **Private Endpoints:** one per **connectivity group** (peered VNets count as one; isolated ones, one each).
- **AMPLS:** normally **1** — groups several PEs and several workspaces.
- Private DNS zones linked to the VNets to resolve the endpoints.

4. Alerts

Signal type	Evaluates	When
Metric	Platform numeric values (CPU, DTU, requestCharge) in near real time	Simple thresholds
Log	Result of a **KQL query** over Log Analytics	Complex conditions: country, event type, grouped counts, non-compliant Policy, sign-in logs
Activity Log	Management events (power off, restart, deallocate, delete)	Resource changes
Resource Health	Resource availability/health	Platform outages
Service Health	Azure incidents	Maintenance/incidents

- **Action Group:** reusable recipients/actions (email, SMS, webhook, Function, Logic App, ITSM). Several rules share one.
- 3 different VM operations (off/restart/deallocate) → **3 rules + 1 action group**.
- Log alert: **period** (window) and **frequency** (how often) are different from the **threshold**.
- **Autoscale** (VMSS): metric + threshold + **duration** (window) + **cooldown**; **multiple profiles** by schedule/date for seasonal traffic. VMSS scales **horizontally**, never vertically.

5. Log Analytics costs

Lever	What it does	Side effect
Commitment tier (100 GB/day, 200…)	Lower price per GB in exchange for a daily commitment	**None**: everything is ingested, alerts intact
Basic Logs (per-table data plan)	Cheaper ingestion	Limited KQL, **no log search alerts**, short retention
Auxiliary Logs	Even cheaper	Basic queries only
Daily cap	Daily ingestion limit	**Data is lost** once exceeded; alerts go blind
Retention	Interactive (up to 2 years) + archive (up to 12 years)	Retention cost

- "Minimize costs, ingest all logs, don't impact the alert" → **commitment tier** (resource: **Workspace1**; the data plan belongs to the table; daily cap discards data).
- 7-year retention of security logs → workspace with archive or export to Storage.

6. Application Insights vs Log Analytics (the distinction that gets tested)

Question	Answer
Correlate resource usage with application data	**Log Analytics** (common repository, KQL)
Simulate users from the Internet, measure transactions, alert on response time	**Application Insights** (availability tests)
See who did what in the portal	**Activity Log**
Dependencies between servers	**Service Map**

7. Decision tree

```
What is the question?
├─ App performance, transactions, synthetics ..... Application Insights
├─ Query/correlate logs, query-based alerts ..... Log Analytics + Log alert
├─ Management audit, shutdowns, changes ..... Activity Log
├─ Network: allowed/denied, next hop, capture ..... Network Watcher
├─ Recommendations without acting ..... Advisor
├─ Act automatically (SQL) ..... Automatic tuning
├─ Collect from hybrid VMs ..... Arc + AMA + DCR + central workspace
├─ Logs over private backbone ..... AMPLS + Private Endpoints per VNet group
└─ Reduce cost without losing data or alerts ..... Commitment tier
```

8. Signals → answer

Wording	Answer
"correlate … with …"	Log Analytics
"simulate user connections from the Internet"	Application Insights
"sign-in logs" + "store and alert"	Log Analytics + signal type Log
"who powered off the VM"	Activity Log alert
"minimum DCRs"	Sets of data types
"minimum private endpoints / AMPLS"	Connectivity groups / 1
"improve SQL performance, minimize admin effort"	Automatic tuning
"packets allowed or denied"	Network Watcher IP flow verify / NSG diagnostics
"120 GB/day, pay-as-you-go, keep all logs, keep alert"	Commitment tier on the workspace
"Basic Logs" on a table with an alert	Breaks the alert → ruled out
"daily cap" + "ingest all"	Loses data → ruled out
"hybrid servers"	Azure Arc + AMA

9. Mistakes to avoid

1. Application Insights to correlate infrastructure and app → Log Analytics.
2. Metric alert for conditions that need content filtering (country, type) → Log.
3. One DCR per region → count data types.
4. One AMPLS per VNet → one is enough.
5. Basic Logs with active alerts.
6. Vertical scaling on VMSS.

Cheatsheet AZ-305 — 08 · Compute and containers

1. The PaaS ↔ IaaS axis: when a VM is required

Requirement in the wording	Answer
Administrative access to the OS, install system-level dependencies	**VM / VMSS** (no App Service, not even Isolated/ASE, gives OS access)
COM component registered on the host	**VM / VMSS**
Legacy software with OS dependencies	**VM / VMSS**
Full .NET Framework + writing temp files + Event Log, minimum maintenance	**App Service (Windows)** is enough
Standard web app/API, minimum maintenance and cost	**App Service**
Not sharing physical hardware with other customers	**Azure Dedicated Hosts** (VMs)
Survive a data center	**Availability Zones** (VMSS across 2+ zones)
Survive a region	Two regions + Traffic Manager / Front Door

2. App Service: plans and ASE

Tier	Dedicated instances	Autoscale / slots	VNet Integration (outbound)	Private Endpoint (inbound)	Isolation
Free / Shared	✗	✗	✗	✗	—
Basic	✓	✗	✓ (v2)	✓	Multi-tenant
Standard	✓	✓	✓	✓	Multi-tenant
Premium (v2/v3)	✓ more CPU/RAM	✓ more slots	✓	✓	Multi-tenant
Isolated (v2)	✓	✓ up to 200 inst.	✓	✓	**Only inside an ASE**

- **App Service Environment (ASE):** full App Service deployment **inside your VNet** (private IP, no public IP on an ILB ASE, all traffic under your NSGs/UDRs/firewall, up to 200 instances). **Still PaaS: no OS access.** Very expensive → only for total isolation, compliance or scale.
- **Private traffic App Service → SQL:** VNet Integration (**delegated**, exclusive subnet) + SQL Private Endpoint (another subnet) + Private DNS zone → **minimum 2 subnets**.
- **Regional redundancy:** a plan lives in one region; multi-region = two deployments + Front Door / Traffic Manager.
- **Deployment slots** (Standard+): swap with no downtime. **Health check** to remove unhealthy instances.
- Repeatable configuration with history → **Bicep/ARM in Azure Repos** (not ZIP, not images).

3. Azure Functions: plans

Plan	Scaling	VNet Integration	Cold start	Cost	When
Consumption	Event-driven, to zero	✗ (limited)	Yes	Minimum	One-off code without private networking
Flex Consumption	Event-driven, to zero	✓	Reduced	Low	Serverless with VNet
Premium	Event-driven, pre-warmed instances	✓	No	Medium	**Access a private IP** + reasonable cost
Dedicated (App Service plan)	Manual/autoscale of the plan	✓	No	Fixed	Reuse an existing plan

- **Durable Functions:** Client (starts) · **Orchestrator** (defines the flow, **durable timers**, conditionals) · Activity (work) · Entity (state).
- **Authorization level:** Anonymous (public) · Function (key) · Admin (master key). Public read-only → **Anonymous + GET only**.
- Queue Storage trigger migrated to AKS → **KEDA + HPA**.

4. Containers: ACI vs Container Apps vs AKS

	Container Instances	Container Apps	AKS
Unit	One container / container group	App with replicas	Full Kubernetes cluster
Automatic scaling	✗	✓ 0→N (HTTP, CPU, **events/KEDA**)	✓ HPA, KEDA, cluster autoscaler
Scale to zero	Only by deleting	✓	With KEDA
Managed ingress / TLS	✗	✓	Ingress controller you deploy
Revisions / canary	✗	✓	With tools (Flux, Istio)
Microservices / Dapr	✗	✓ built in	Add-on
Cluster control	—	Hidden	**Full** (kubectl, Helm, CRDs, networking)
Startup	Seconds	Seconds	Minutes (nodes)
Billing	Per second of CPU/RAM	Per consumption	Nodes (VMs)
Use case	Jobs, one-off tasks, **AKS burst (virtual nodes)**	APIs, web apps, microservices without managing K8s	Kubernetes workloads, fine-grained control

- **Service Fabric:** Microsoft's microservices orchestrator, **hybrid on-prem + Azure**, independent updates, self-healing, hyperscale.
- **Azure Batch:** parallel compute with its own scheduler. **CycleCloud:** HPC clusters with a **third-party scheduler** (Slurm, PBS, LSF).

- **Container Registry Premium: geo-replication** of images for multi-region AKS; CMK; private link.

5. AKS: pieces that appear in the exam

Piece	What it does	Signal
Ingress controller (NGINX, AGIC)	Single L7 entry point: paths, hosts, TLS, **mTLS**, non-standard ports; avoids one LB per service	"expose multiple APIs", "non-default ports", "mTLS", "min cost"
Service LoadBalancer	One Azure LB per service, L4	Only if a single service
ExternalName service	DNS alias to something external	Distractor
HPA	Scales pods by CPU/RAM or external metrics	With KEDA
KEDA	Scales pods by **events** (queues, Service Bus, Event Hubs), 0↔1; creates the HPA	"same scaling as Functions Consumption"
Cluster autoscaler	Scales **nodes**; works with kubenet and Azure CNI	"nodes", "capacity"
Virtual nodes / Virtual Kubelet	Pods on ACI for burst; **Azure CNI only**	Ruled out if they say "kubenet"
Dapr	Building blocks: **state management**, pub/sub, service invocation, secrets	"state management"
Flux	GitOps: syncs the cluster with a repo	"deploy from Git"
Istio	Service mesh: mTLS between pods, canary, traffic	"traffic between services"
Azure Policy add-on	Gatekeeper/OPA (EnforceOPAConstraint / Rego)	"pod governance"
kubenet vs Azure CNI	kubenet: pod IPs NATed; CNI: VNet IPs (more IPs, direct integration)	"pods reachable from VNet" → CNI

- Front Door → 2 AKS clusters in 2 regions with centralized SSL.
- APIM + AKS same VNet, private IP, mTLS, rate limit (APIM options) → Premium Internal.

6. VMs and VMSS: exam rules

Topic	Rule
SLA	Single VM with Premium SSD 99.9 % · Availability Set 99.95 % · Availability Zones 99.99 %
Availability Set	Fault / update domains, same datacenter; does not protect against datacenter failure
Availability Zones	Separate datacenters within the region; "data center failure"
VMSS	**Horizontal** scaling; autoscale with metrics, duration, cooldown, **profiles** by schedule/date; software at provisioning with **Custom Script / DSC extension** in extensionProfile + uploaded script
Series	D/Ds general purpose (default and cost), E memory, F compute, L local disk, M massive memory, N GPU (NC compute, NV visual), H HPC; **"s" = Premium storage**; SR-IOV = Accelerated Networking
Disks	OS: Premium SSD **v1** (v2 not valid for OS); SQL: log **None**, data **ReadOnly**
Dedicated Hosts	Non-shared physical hardware
Confidential VMs	Data in use encrypted (TEE)
Scale with no downtime	Horizontal + VMSS + Availability Set/Zones
JIT VM access	Defender for Cloud opens management ports temporarily (it is not PIM)
VMware migration	Azure Migrate appliance (discovery) / AVS only if they want to stay on VMware

7. Decision tree

```
Which hosting?
├─ Needs OS / COM / host dependencies ..... VM / VMSS (zones for datacenter)
├─ Standard web/API, min maintenance ..... App Service (Standard/Premium)
├─ Same but no public IP / total isolation ..... ASE (still no OS)
├─ Event-driven code ..... Functions (Premium if private IP)
├─ One-off container / job ..... Container Instances
├─ Microservices without managing K8s ..... Container Apps
├─ Kubernetes with control ..... AKS
├─ Hybrid on-prem + Azure microservices ..... Service Fabric
├─ HPC with third-party scheduler ..... CycleCloud
└─ Managed parallel compute ..... Batch
```

8. Signals → answer

Wording	Answer
"administrators access to the OS"	VMs / VMSS (never App Service)
"custom COM component" + "data center unavailable" + "min cost"	VMSS + LB across 2 availability zones
"Isolated plan" + "redundancy if region fails"	Does not meet it: one region, no OS
".NET web service, temp files, event log, min cost"	App Service web app
"Event Grid + C# + private IP + min cost"	Functions Premium
"state management" (AKS)	Dapr

Wording	Answer
"same scaling as Consumption" (AKS, kubenet)	KEDA + HPA
"expose APIs on non-default ports + mTLS + min cost"	Ingress controller
"image replicated to AKS in multiple regions"	Container Registry Premium (geo-replication)
"HPC + third-party scheduler"	CycleCloud
"on-prem + Azure microservices, self-healing"	Service Fabric
"define config of 50 App Services + version history"	Bicep in Azure Repos
"15,000 IOPS + SR-IOV + min cost"	DS series

9. Mistakes to avoid

1. ASE/Isolated as the answer to "OS access".
2. Two regions when they ask to survive a data center and minimize cost → zones.
3. Virtual nodes with kubenet.
4. HPA alone (CPU) to replicate Functions Consumption → KEDA + HPA.
5. Container Instances for an API with autoscaling → Container Apps.
6. Vertical scaling on VMSS.

Cheatsheet AZ-305 — 09 · Azure SQL, Cosmos DB and data platforms

1. Azure SQL products

Product	What it is	Choose when	Rule out when
Single Database	One isolated DB on a logical server	Independent DB; **Serverless**; modern features	Cross-database queries, three-part naming, SQL Agent
Elastic Pool	Several single DBs sharing compute	Many DBs with variable usage; **reserved capacity**; Hybrid Benefit; autoscaling across DBs	They ask for Serverless (does not exist in a pool)
Managed Instance	Instance almost 100 % compatible with SQL Server	**Cross-database**, linked servers, SQL Agent, three-part naming, **server-side transactions**, lift-and-shift migration	They ask for Serverless or the minimum cost without instance features
SQL Server on VM	IaaS	OS control, full compatibility, features MI does not cover	"minimize administrative effort"

- Two DBs + "server-side transactions across DB1 and DB2" + "minimize admin effort" → **MI**.
- "server.database.table" → **MI or VM** (two correct); never Single DB.
- "automatically scales compute + per-second billing" → **Single DB, General Purpose Serverless**.
- "20 DBs, varying usage, 99.99 %, dynamic compute, **reserved capacity**" → **Elastic pool** (Serverless does not support reservations).
- "License Mobility / Software Assurance" → **Azure Hybrid Benefit** → **vCore** model (pool or MI).

2. Purchasing models

	DTU	vCore
Unit	CPU+RAM+IO bundle (DTU)	vCores + RAM + storage separately
Tiers	Basic → Standard → Premium	General Purpose → Business Critical → Hyperscale
Hybrid Benefit	✗	✓
Reserved capacity	✗	✓
Serverless	✗	✓ (GP only, single DB)
Managed Instance	✗	✓
Status	Legacy, small DBs	Recommended

3. Tiers (DTU ↔ vCore translation)

Tier	Model	Equivalent to	Storage	I/O latency	HA	Zone-redundant	Max size	Serverless
Basic	DTU	—	Remote	High	Basic	✗	**2 GB**	✗
Standard	DTU	General Purpose	Remote	Medium	Compute reassignment	✗	1 TB	✗
General Purpose	vCore	Standard	Remote (Premium Storage)	Medium	Compute reassignment	✓ optional	4 TB	✓
Premium	DTU	Business Critical	**Local SSD**	**Minimum**	Primary + **synchronous** secondaries	✓	4 TB	✗
Business Critical	vCore	Premium	**Local SSD**	**Minimum**	Primary + 3 synchronous secondaries	✓ (survives 2 zones)	4 TB	✗
Hyperscale	vCore	—	Page servers (decoupled)	Low-medium	Optional HA replicas, **named replicas** for reads	✓	**100 TB**	✗

- "minimize I/O latency" + "2 zones fail" + "automatic failover" → **Business Critical** (or Premium in DTU).
- "failover without data loss" + "zone outage" + "min cost" (DTU options) → **Premium** (Standard has no zone redundancy).
- "expand to 50 GB, minimize cost" → **Standard**, not Premium.
- "> 4 TB" / "50 TB relational" / "200-300 concurrent reads" → **Hyperscale** (Synapse dedicated pool = **128 concurrent queries** max).
- "improve performance, minimize admin effort" → **Automatic tuning** (acts) vs Advisor/QPI/Monitor (only report).

4. Security in Azure SQL

Need	Solution
File encryption at rest	**TDE** (with CMK in Key Vault if "manage own keys"; Policy DeployIfNotExists to enable it)
Column never in clear text, only the client decrypts	**Always Encrypted**
Some see everything, others partially	**Dynamic Data Masking**
Firewall that travels with the failover	**Database-level** rules
Entra ID users without keys	Entra ID authentication + managed identity
Auditing / threats	SQL Auditing, Defender for SQL

5. Cosmos DB

Topic	Rule
APIs	**NoSQL** (JSON documents, SQL-like queries, **non-relational**) · **PostgreSQL** (relational, standard SQL, distributed, geo-replicas) · MongoDB · Cassandra (wide-column, CQL) · Gremlin (graphs) · Table
"store and access relationally" + "SQL queries"	**PostgreSQL** (NoSQL fails on "relational")
Multi-region writes	Only service with local writes in several regions; "read and write in local region" → Cosmos DB
Consistency levels	Strong · Bounded staleness · Session · Consistent prefix · Eventual
Partition key	**Immutable**; changing it = **create a new container** and migrate
Data RBAC	Built-in Data Reader / Contributor with Entra ID identities, no account keys or resource tokens
Analytics without ETL	**Synapse Link** → columnar analytical store, near real time, without consuming RUs
RU alerts	Log Analytics query over requestCharge_s
Catalog with variable schema + global distribution	Cosmos DB

6. Other data platforms

Service	What for	Signal
Synapse dedicated SQL pool	MPP warehouse for heavy analytical queries	"data warehouse" with **≤128 concurrent**
Synapse serverless SQL	Query files in the lake without loading	"query data directly in the lake"
Data Lake Storage Gen2	Lake with directories by date (Event Hubs Capture), POSIX ACLs	"organize by date/time", "Hadoop"
Databricks	Spark; **Credential Passthrough** (user identity → ACLs) or Unity Catalog; on-prem app auth with **service principal**	"user identity for folder ACLs"
Data Factory / Synapse pipelines	Low-code ETL/ELT; **Self-hosted IR** for on-prem; **Azure-SSIS IR** for SSIS packages (SSISDB in **Azure SQL Database**)	"transform during load", "SSIS packages"
Data Share	Share snapshots with another organization	"partner", "restricted access to snapshots"
Purview	Catalog, classification, lineage	"discover sensitive data"
Azure Database for MySQL/PostgreSQL Flexible	Regional DR with **promoted read replica** (min downtime); geo-backup = geo-restore (hours)	"minimize downtime to paired region" → read replica
Table Storage	Independent tables over REST, minimum cost, GRS	"REST + independent tables + second region + min cost"
Azure Cache for Redis	Session state / cache next to the app	"session state", "serializable"

7. Decision tree (Azure SQL)

```
Instance or DB?
├─ Cross-database / Agent / linked servers ..... Managed Instance (VM if you need the OS)
├─ Many DBs with variable usage / reservations . Elastic pool
└─ Isolated DB
    ├─ Autoscaling + per second ................ General Purpose Serverless
    ├─ Minimum latency / 2 zones / sync ........ Business Critical (Premium in DTU)
    ├─ > 4 TB / high concurrency ............... Hyperscale
    ├─ < 2 GB, minimum cost, no HA ............. Basic
    └─ General ................................. General Purpose (Standard in DTU)
```

8. Signals → answer

Wording	Answer
"server-side transactions across DB1 and DB2" + min effort	Managed Instance
"server, database, and table name"	MI + SQL on VM
"per second billing"	Single DB, General Purpose Serverless
"reserved capacity" + many DBs	Elastic pool
"Software Assurance"	vCore + Hybrid Benefit
"minimize I/O latency" + "two zones"	Business Critical
"50 TB relational" + "200-300 concurrent reads"	Hyperscale
"read and write in the local region"	Cosmos DB (multi-region writes)
"relationally" + Cosmos API	PostgreSQL
"change partition key, first step"	Create a new container
"Entra users read access to Cosmos data, no keys"	Built-in Data Reader (RBAC)
"near real-time analytics on Cosmos DB, no ETL"	Synapse Link
"SSISDB catalog, no hybrid connectivity, min cost"	Azure SQL Database
"MySQL flexible, minimize downtime to paired region"	Read replica

9. Mistakes to avoid

1. Serverless in an elastic pool or MI → does not exist.
2. Reserved capacity with Serverless → incompatible.
3. Standard (DTU) for zone redundancy → it does not have it.
4. Synapse dedicated pool with >128 concurrent reads → Hyperscale.
5. Cosmos NoSQL as "relational" because it has SQL queries.
6. Premium for 50 GB when Standard is enough and they ask for minimum cost.
7. Geo-restore/geo-backup when they ask to minimize downtime.

Cheatsheet AZ-305 — 10 · Backup, business continuity, DR and hybrid networking

1. Backup vs Site Recovery

	Azure Backup	Azure Site Recovery (ASR)
What it is	Backup copies in a Recovery Services vault	Continuous VM replication and orchestrated failover
Protects against	Corruption, deletion, ransomware, data loss	Datacenter / region outage (continuity)
RPO	Hours (backup frequency)	Minutes
RTO	Hours (restore)	Minutes (Recovery Plan)
Cost	Low	Low (no active VM in the target until failover)
When	"recover the data", "retain N years"	"remain available", "relaxed RPO + min cost", VM migration

- **SQL on VM, nightly batch, RPO 24 h, RTO 15 min, regional, automated, min cost → ASR** (a multi-region AG would be expensive; backup does not meet the RTO).
- **Changing vault:** a VM can only be in one vault → first **Stop backup** in the source vault.
- **The VM's Disaster recovery blade = ASR**, not Backup.

2. Azure Backup agents and components

Component	What it protects	When
VM backup extension	Entire Azure VMs	VM already in Azure; no agent to install
MARS agent	Files and folders on Windows servers (on-prem or Azure)	"specific files/folders", simplest solution
MABS (Azure Backup Server = DPM for Azure)	Full applications: SQL Server, Exchange, SharePoint, Hyper-V/VMware VMs, bare-metal	"application-aware", "centralized backup of many on-prem servers"
DPM (System Center)	Same as MABS, with a System Center license	They already have System Center
SQL backup on Azure VM	SQL databases with log backups every 15 min	RPO 15 min for SQL on VM
Backup of Azure Files / Blobs / Disks / PaaS DBs	PaaS resources	Depends on the resource

- Long retention (7 years, two separate locations) → vault with **GRS** + long-term retention; historical alternative DPM + tape.
- **Soft delete for backups** (14 days) and **Multi-user authorization with Resource Guard** against ransomware.

3. Resource Guard and roles

Element	Detail
Resource Guard	Resource in **another subscription/tenant**; critical vault operations (disable soft delete, reduce retention, stop backup) require a role on it → anti-ransomware separation of duties
Contributor	Create and manage the vault
Site Recovery Contributor	Enable replication, manage jobs (does not create the vault)
Site Recovery Operator	Run failover / failback
Backup Contributor / Operator / Reader	Equivalents for Backup

4. Storage: continuity

- **GRS/RA-GRS** for region; **ZRS** for datacenter; failover leaves the account in **LRS** → reconfigure GRS for failback.
- **Azure File Sync** for continuous file access when a site goes down (it is not backup).
- **Immutable storage** protects against deletion/modification, it does not restore.

5. Hybrid networking

Technology	Medium	Encryption	Bandwidth / SLA	When
Site-to-Site VPN	Internet	IPsec/IKE	Up to ~10 Gbps (SKU), no network SLA	Offices ↔ Azure; **ExpressRoute failover** ("must use Internet, no MPLS")
Point-to-Site VPN	Internet	SSTP/IKEv2/OpenVPN	Per client	Remote users; certificates (public root is uploaded, client stays) or Entra ID
ExpressRoute	Private circuit (provider, MPLS)	**Not encrypted by default**; add IPsec over ER or **MACsec** (ER Direct)	Up to 100 Gbps, SLA 99.95 %, **redundant pair of cross connections**	Private, predictable, high volume
ExpressRoute Global Reach	Microsoft backbone	—	—	Connect sites to each other via ER
VNet peering	Backbone	—	—	VNet ↔ VNet (regional/global); **not transitive**; "Use remote gateways" to share a gateway

Technology	Medium	Encryption	Bandwidth / SLA	When
Virtual WAN	Managed hubs	—	—	Multi-region, transitive, automatic P2S, ER (Standard)
Azure Bastion	—	TLS	—	RDP/SSH without public IP
NAT Gateway	—	—	—	Outbound Internet with a fixed IP

- **Gateway Subnet** required for VPN/ER gateways; Local Network Gateway represents the on-prem network.
- **Zone-redundant gateways** (AZ SKUs) for HA; VPN Gateway **active-active** for redundancy.
- **Subnet sizing:** Azure reserves **5 IPs**; /27 = 27 usable, /26 = 59, /25 = 123, /24 = 251. Never overlap with on-prem.
- **UDR with NVA/firewall as next hop** forces inspection; a single NVA = single point of failure → pair of NVAs + ILB or Azure Firewall.
- **DNS:** Private DNS Zones for multi-VNet and private endpoints; privatelink.* zone per service.
- **Private Endpoint** (private IP for the service, blocks public access) vs **Service Endpoint** (subnet → service over the backbone, public IP is kept).

6. Network security

Service	Layer	Filters by	When
NSG	4	IP, port, protocol, service tags, ASGs	Subnet / NIC
Application Security Groups	4	Group NICs for NSG rules	Simplify NSGs
Azure Firewall	4-7	IP, port, **FQDN**, tags, threat intel; Premium: TLS inspection, IDPS	Central hub, FQDN filtering
WAF (App Gateway / Front Door)	7	OWASP, bots, rate limit	Web apps
DDoS Protection	3-4	Volumetric	Public IPs
JIT VM access (Defender)	—	Opens management ports temporarily	RDP/SSH on demand
Network Watcher	—	Diagnostics (IP flow verify, next hop, flow logs)	"allowed or denied"

7. Disaster recovery per service (cross summary)

Service	DR mechanism with minimum downtime	Cheap mechanism (data)
VM	ASR	Backup
SQL on VM	Asynchronous AG / Distributed AG	Backup + log shipping
Azure SQL DB	Auto-failover group	Geo-restore
Managed Instance	Auto-failover group	Geo-restore
MySQL/PostgreSQL Flexible	Promoted read replica	Geo-redundant backup
Cosmos DB	Multi-region (writes or automatic failover)	Continuous backup
Storage	RA-GRS / account failover	GRS + restore
App Service / AKS	Two regions + Front Door / Traffic Manager	Redeploy from IaC
Key Vault	Automatically replicated to the paired region	—

8. Decision tree

```
What must survive and with what RPO/RTO?
├─ Data only, hours of RTO, min cost ............. Backup (GRS vault, retention)
├─ VM available in another region, RPO minutes ... ASR + Recovery Plan
├─ Files/folders of a server ..................... MARS
├─ SQL/Exchange/Hyper-V on-prem centralized ....... MABS
├─ Backups resistant to a compromised admin ...... Resource Guard in another subscription
├─ Private on-prem connection, SLA, high volume .. ExpressRoute (+ IPsec if encryption is required)
├─ ExpressRoute failover over the Internet ....... Site-to-Site VPN
├─ New offices with no special requirements ...... Site-to-Site VPN
├─ VNet ↔ VNet ................................... Peering (Virtual WAN if transitivity/multi-region)
└─ Traffic inspected by FQDN ..................... Azure Firewall (secured hub) + UDRs
```

9. Signals → answer

Wording	Answer
"RPO 24 hours, RTO 15 min, regional, automated, min cost" (SQL VM)	Azure Site Recovery
"use RSV2 to protect VM2 (in RSV1), first step"	Stop backup in RSV1
"files and folders", "simplest"	MARS
"SQL Server / Exchange / SharePoint on-prem"	MABS
"ransomware" + "role in Sub2 first"	Resource Guard
"create and manage the vault"	Contributor
"ExpressRoute fails, use Internet, no MPLS"	Site-to-Site VPN

Wording	Answer
"private, redundant pair of cross connections"	ExpressRoute
"offices in two cities, new subscription"	Site-to-Site VPN
"encryption required over ExpressRoute"	IPsec over ER / MACsec
"30 VMs, on-prem 172.16.0.0/16, S2S VPN"	192.168.0.0/24
"force traffic through firewall between VNets"	UDRs with the firewall as next hop
"packets allowed or denied"	Network Watcher
"RDP/SSH without public IP"	Bastion
"temporary management ports"	JIT VM access

10. Mistakes to avoid

1. Backup when an RTO of minutes is required → ASR.
2. Configuring RSV2 without first stopping the backup in RSV1.
3. Confusing the "Disaster recovery" section (ASR) with Backup.
4. Assuming ExpressRoute encrypts.
5. A second ExpressRoute as failover when MPLS is forbidden → VPN.
6. Forgetting the 5 reserved IPs when sizing subnets.
7. Treating peering as transitive.

Cheatsheet AZ-305 — 11 · Costs, purchasing models and composite SLAs

1. Compute savings models

Mechanism	What it is	Savings	Commitment	Applies to	When
Pay-as-you-go	Per second/hour with no commitment	0 %	None	Everything	Unpredictable or temporary workloads
Reserved Instances (RI)	1- or 3-year reservation of **one VM family in one region**	Up to ~72 %	High: fixed family + region (size flexibility within the family)	VMs, SQL DB/MI vCore, Cosmos DB, Storage, App Service, Synapse, Redis…	**Stable and predictable** workload on specific compute
Savings Plan for compute	**Hourly spend** commitment (€/h) for 1 or 3 years	Up to ~65 %	Medium: **any region, family or compute service**	VMs, App Service, Functions Premium, Container Apps, Container Instances, Dedicated Hosts	Stable workload that **changes in type or region**
Spot VMs	Spare capacity, **evictable** with 30 s notice	Up to ~90 %	None	VMs, VMSS, AKS node pools	**Interruptible** workloads: batch, render, test, CI; never critical production
Azure Hybrid Benefit	Use **Windows Server / SQL Server licenses with Software Assurance** (or subscription)	Up to ~40 % Windows, ~55 % SQL (85 % combined with RI)	Own licenses	Windows VMs, SQL on VM, Azure SQL DB/MI **vCore**, AKS Windows nodes, Azure Stack HCI	"Software Assurance", "License Mobility", "existing licenses"
Dev/Test pricing	Reduced rates with no Windows license in Dev/Test subscriptions (Visual Studio)	Variable	—	Non-production environments	"development and test environment"
Autoscale / Serverless / scale to zero	Pay only for what is used	Variable	—	VMSS, App Service, Functions, Container Apps, SQL Serverless	Variable workload with troughs
Scheduled shutdown	Automation runbook / VM Auto-shutdown	100 % of the time switched off	—	Dev/test VMs	Business hours

- RI and Savings Plan **combine** (RI is applied first). Both are **billing** constructs, they do not reserve physical capacity (for that: **capacity reservations** or Dedicated Hosts).
- **Reserved capacity** does not exist with **Serverless** (SQL) or with Spot.
- "Minimize costs" in an exam question means **the cheapest option that meets the technical requirements**, never the cheapest in absolute terms.

2. Storage and data savings

Lever	Effect
Access tiers Hot → Cool → Cold → Archive + lifecycle management	Lower cost per GB, higher access cost/latency
LRS < ZRS < GRS < RA-GRS < GZRS < RA-GZRS	Fewer copies = lower cost; pick the minimum that protects what is required
Standard vs Premium	Premium only if latency/IOPS are required
Storage reserved capacity (100 TB / 1 PB, 1-3 years)	Discount on Blob/ADLS
SQL: elastic pool (share), Serverless (pause), reservations, Hybrid Benefit, Standard vs Premium	See cheatsheet 09
Log Analytics: commitment tiers, Basic/Auxiliary Logs, retention, daily cap	See cheatsheet 07
Managed disks: Standard HDD < Standard SSD < Premium SSD < Premium v2 < Ultra	Pick by required IOPS/latency

3. Cost Management + Billing

Tool	What it does	Signal
Cost analysis	Explore and group costs (by tag, RG, service, subscription)	"review weekly costs", "group by department" → **tags**
Budgets	Spending threshold with alerts (email / action group); does not block spending	"notify when spend exceeds"
Cost alerts	Budget, anomalies, credit	"unexpected increase" → anomaly alerts
Advisor (Cost)	Underutilized VMs, orphaned disks, recommended RIs; **does not act**	"recommendations"
Pricing calculator / TCO calculator	Estimate the cost of an architecture / compare with on-prem	"estimate before migrating"
Exports	Scheduled dump of costs to Storage	"analyze in Power BI"
Management groups + subscriptions	Billing structure (EA: Enrollment → Departments → Accounts → Subscriptions; MCA: Billing profiles → Invoice sections)	"chargeback"
Mandatory tags (Policy)	Guarantee cost dimensions	"ensure every resource has a cost center"
Cost Management Reader / Contributor	RBAC to view / manage budgets	"auditors must review costs" → Reader

4. Service SLAs (the ones you must know)

Service	SLA
Single VM with Premium SSD / Ultra	99.9 %
Single VM with Standard SSD	99.5 %
Single VM with Standard HDD	95 %
VMs in an **Availability Set**	99.95 %
VMs in **Availability Zones** (≥2)	99.99 %
App Service (Basic+)	99.95 %
Azure SQL Database / MI	99.99 % (99.995 % Business Critical zone-redundant)
Cosmos DB	99.99 % single-region; 99.999 % multi-region reads
Storage (LRS/ZRS/GRS)	99.9 % read/write (RA-GRS read 99.99 %)
Load Balancer Standard	99.99 %
Application Gateway v2	99.95 %
Front Door	99.99 %
Traffic Manager	99.99 %
VPN Gateway	99.9 % (Basic) / 99.95 % (Standard+)
ExpressRoute	99.95 %
AKS (control plane, with uptime SLA)	99.95 % (99.9 % without zones)
Key Vault	99.99 %
Functions Premium / App Service	99.95 %

5. Composite SLA calculation

In series (all must work): multiply.

```
Web (99.95 %) → SQL (99.99 %)  =  0.9995 × 0.9999 = 0.9994 → 99.94 %
```

Each component in series **lowers** the total SLA. One component at 99.9 % drags the whole thing below 99.9 %. **In parallel (one is enough):** 1 − product of the unavailabilities.

```
Two independent regions at 99.95 %:  1 − (0.0005 × 0.0005) = 0.99999975 → 99.99997 %
```

Adding redundancy **raises** the SLA above the individual component. **Mixes:** first compute the parallel blocks, then multiply the series.

```
Traffic Manager (99.99 %) → [Region A ‖ Region B], each region = App (99.95 %) × SQL (99.99 %) = 99.94 %
Parallel: 1 − (0.0006)² = 99.99996 %
Total: 0.9999 × 0.9999996 ≈ 99.99 %
```

Exam rules:

- "SLA 99.99 % per tier" with VMs → **Availability Zones** (Availability Set only gives 99.95 %).
- Adding a load balancer or gateway in series **lowers** the total; it must be compensated with parallelism.
- Minutes of unavailability per month: 99.9 % ≈ 43 min · 99.95 % ≈ 22 min · 99.99 % ≈ 4.3 min · 99.999 % ≈ 26 s.
- The SLA of the weakest component in series is the ceiling of the total.

6. Decision tree (compute savings)

```
What is the workload like?
├─ Stable, fixed compute, same region ..... Reserved Instances (+ Hybrid Benefit if there are licenses)
├─ Stable but changes family/region ..... Savings Plan
├─ Interruptible (batch, test, render) ..... Spot
├─ Variable with troughs ..... Autoscale / Serverless / scale to zero
├─ Dev/test during business hours ..... Auto-shutdown + Dev/Test pricing
└─ I have licenses with Software Assurance ..... Hybrid Benefit (vCore in SQL)
```

7. Signals → answer

Wording	Answer
"predictable workload, 3 years, same VM size"	Reserved Instances
"commit to spend, flexibility across regions and VM families"	Savings Plan
"batch job can be interrupted, lowest cost"	Spot VMs
"Software Assurance", "existing licenses"	Azure Hybrid Benefit
"auditors review costs weekly"	Cost Management (Cost Management Reader)
"alert when spending exceeds"	Budget

Wording	Answer
"group costs by department"	Tags + Cost analysis
"SLA 99.99 % for VMs"	Availability Zones
"estimate cost of migration vs on-prem"	TCO calculator
"Serverless" + "reserved capacity"	Incompatible → elastic pool

8. Mistakes to avoid

1. Spot for production or non-interruptible workloads.
2. RI when the workload changes family/region → Savings Plan.
3. Hybrid Benefit in the DTU model → vCore only.
4. Availability Set when 99.99 % is required.
5. Adding SLAs instead of multiplying them in series.
6. Choosing the cheapest option that does not meet the technical requirement.

Cheatsheet AZ-305 — 12 · Azure Migrate and migration strategies

1. The 5 Rs (strategies)

Strategy	What it is	Typical target	Signal
Rehost (lift and shift)	Move as is, no changes	VMs, AVS	"minimize changes", "quickly"
Refactor (replatform)	Minimal changes for PaaS	App Service, MI, Container Apps	"minimize code changes, use PaaS"
Rearchitect	Redesign (microservices, serverless)	AKS, Functions, Cosmos DB	"modernize", "cloud-native"
Rebuild	Rewrite from scratch	PaaS/serverless	"new application"
Replace	Replace with SaaS	Microsoft 365, Dynamics	"SaaS alternative"

2. Azure Migrate: components

Component	What it does	Exam detail
Azure Migrate project (hub)	Resource in Azure that centralizes discovery, assessment and migration; integrates Microsoft and third-party tools	Created first; discovers nothing by itself
Azure Migrate appliance	VM (OVA for VMware, VHD for Hyper-V, installer for physical) deployed **on-prem**; connects to vCenter / Hyper-V hosts / servers via WinRM-SSH; discovers inventory, **performance** and **dependencies** (agentless)	"what should you configure for the cluster" → appliance
Discovery	Inventory of VMs, SQL Server, web apps (IIS, Tomcat), apps, roles	Continuous while the appliance is active
Assessment	Recommends VM SKU (as-is or **performance-based**), disk, monthly cost, readiness (ready / conditionally / not ready)	"identify resources required", "right-size" → performance-based
Dependency analysis	Agentless (via appliance) or agent-based (Log Analytics + agents); map of what talks to what	"group servers to migrate together", "waves"
Business case	TCO on-prem vs Azure	"justify migration"
Migration and modernization (formerly Server Migration)	Replicates and migrates VMware VMs (agentless or agent-based), Hyper-V, physical, other clouds; **test migration** without affecting production	"test the migration before cutover"
Azure Migrate: Database (DMS)	SQL Server assessment and migration	See section 4
Web app assessment / migration	.NET and Java apps to **App Service** or AKS	"migrate IIS web apps to App Service"
Data Box	Offline migration of massive data	"TB with limited bandwidth"

- Flow: **project → appliance → discovery → assessment (+dependencies) → replicate → test migration → migrate (cutover)**.
- **Azure VMware Solution (AVS):** VMware private cloud **inside Azure** (dedicated hosts); to keep using vSphere/vCenter without converting. Move tool: **HCX**. Choose AVS only if the wording asks to keep VMware.
- **Azure Migrate vs ASR for migrating VMs:** Migrate is the current and recommended tool (uses ASR underneath for VMware agent-based / physical); direct ASR is valid for DR and simple VM migrations.

3. Server migration: which tool

Source	Tool	Mode
VMware vSphere	Azure Migrate (agentless, via appliance)	Snapshot replication; no agent on the VMs
VMware without vCenter access / physical / AWS / GCP	Azure Migrate agent-based (Mobility agent + replication appliance)	Agent on each server
Hyper-V	Azure Migrate (agentless, Hyper-V replication provider on hosts)	No agent on VMs
AWS EC2 / GCP	Azure Migrate agent-based (treated as physical)	Agent
VMware keeping VMware	AVS + HCX	vMotion / bulk migration
DR only or very simple migration	Azure Site Recovery	Continuous replication

4. Database migration

Source → Target	Tool	Downtime	Notes
Assess SQL Server beforehand	**Data Migration Assistant (DMA)** or Azure Migrate SQL assessment	—	Compatibility, deprecated features, SKU recommendation
SQL Server → Azure SQL Database	**BACPAC** (SSMS/sqlpackage) · DMS · **Azure Data Studio + SQL Migration extension**	BACPAC high; DMS/ADS online low	Single DB does not support cross-database
SQL Server → Managed Instance	**Azure Data Studio + SQL Migration ext.** (online/offline) · **DMS** (online/offline, many DBs) · **Log Replay Service** (native backups) · **Managed Instance link** (distributed AG with MI, near-zero downtime)	Minimal with online	50 DBs offline with minimal effort → DMS

Source → Target	Tool	Downtime	Notes
SQL Server → SQL on VM	**Distributed AG** (from an existing AG, cross-version, Windows→Linux) · Log shipping · Backup/restore · ASR (whole VM)	Distributed AG minimal	See SQL VM cheatsheet
Oracle / MySQL / DB2 / SAP ASE / Access → SQL	**SQL Server Migration Assistant (SSMA)**	High	Converts schema and code
MySQL / PostgreSQL → Azure Database Flexible Server	**DMS** (online with replication) · dump/restore · **read replica** from on-prem does not apply	Minimal with DMS online	
MongoDB → Cosmos DB for MongoDB	DMS · Cosmos DB migration tool · mongodump/mongorestore	Variable	
Cassandra → Cosmos DB for Cassandra	Dual-write proxy · Spark		
Massive data (TB) to Storage/ADLS	AzCopy · Data Factory · **Data Box** / **Import-Export**		See cheatsheet 03
SSIS packages	**Azure-SSIS Integration Runtime** in ADF; SSISDB in Azure SQL Database (cost) or MI (compatibility)		No changes to packages
Continuous on-prem ↔ Azure access (not migration)	**Self-hosted Integration Runtime** (formerly Data Management Gateway)		ADF/Synapse with on-prem sources

- **The words "minimize downtime"** → rule out BACPAC, backup/restore, copying VHDs; look for continuous replication (DMS online, ADS online, Distributed AG, MI link, ASR).
- **The word "offline"** → DMS offline or backup/restore; the criterion becomes effort and number of DBs.

5. Application migration

Scenario	Target	Tool / pattern
IIS .NET web app with no OS access required	App Service (Windows)	Azure Migrate web app migration / App Service Migration Assistant
App with COM components, OS dependencies	VM / VMSS	Rehost
App with LDAP against AD DS, no VPN allowed	VM + **Entra Domain Services**	Identity refactor
On-prem app with IWA/Kerberos for remote users	Stays on-prem + **Application Proxy (KCD)**	No migration
Microservices in containers	AKS / Container Apps	Rearchitect
Functions Consumption with queue trigger → AKS	AKS + **KEDA + HPA**	Replatform
Java Spring apps	Azure Spring Apps / Container Apps	Refactor
In-memory session state → scalable	App Service + **Azure Cache for Redis**	Refactor

6. File and File Server migration

Scenario	Solution
On-prem file server, keep local + cloud access	**Azure File Sync** (cloud tiering)
Migrate shares to Azure Files	Azure File Sync · Storage Migration Service (Windows Admin Center) · Robocopy / AzCopy
Massive data, little bandwidth	**Data Box** (Microsoft ships it) / **Import-Export** (your disks)
Continuous transfer from on-prem	Data Box Gateway
Rarely accessed files, minimum cost	Blob **Archive** (24 h rehydration acceptable)

7. Decision tree

```
What am I migrating?
├─ VMware / Hyper-V / physical VMs ....... Azure Migrate (appliance → assessment → migrate); AVS if they want to stay on VMware
├─ Only assess SQL Server ................ DMA / Azure Migrate SQL assessment
├─ SQL → Single DB ....................... BACPAC (downtime OK) / ADS-DMS online (min downtime)
├─ SQL → Managed Instance ................ ADS + SQL Migration ext. (online) / DMS (many DBs, offline) / MI link
├─ SQL → SQL on VM from an AG ............ Distributed AG
├─ Oracle/MySQL/DB2 → SQL ................ SSMA
├─ MySQL/PostgreSQL → Flexible Server .... DMS online
├─ SSIS ................................ Azure-SSIS IR + SSISDB in Azure SQL DB
├─ IIS web app ......................... App Service (Migrate web app tool)
├─ Files, TB, little bandwidth ........... Data Box / Import-Export
└─ Continuous access, not migration ...... Self-hosted IR / On-premises Data Gateway
```

8. Signals → answer

Wording	Answer
"VMware cluster, identify resources required, minimize effort"	Azure Migrate appliance (project as hub)
"keep using vSphere / VMware tools"	Azure VMware Solution

Wording	Answer
"group servers by dependencies"	Dependency analysis
"right-size based on actual usage"	Performance-based assessment
"test before cutover"	Test migration
"50 databases, MI, offline, min effort"	Database Migration Service
"SQL 2008 → MI, minimize downtime"	Azure Data Studio (SQL Migration extension)
"existing AG → SQL 2019 on Linux VM, minimize downtime"	Distributed availability group
"Oracle to Azure SQL"	SSMA
"SSIS packages, no hybrid connectivity, min cost"	Azure-SSIS IR + SSISDB in Azure SQL Database
"5 TB rarely accessed files"	Blob Archive
"10 TB, limited bandwidth"	Data Box
"file server with local cache"	Azure File Sync

9. Mistakes to avoid

1. Azure Migrate as "the tool that migrates databases" → it is the hub; DMS/ADS do the migrating.
2. DMA to migrate to MI → DMA only assesses (and migrates to Single DB).
3. SSMA for SQL Server → SQL → SSMA is for **other engines**.
4. AVS when native Azure VMs are required.
5. BACPAC with "minimize downtime".
6. Copying VHDs when ASR / Azure Migrate exist.

Cheatsheet AZ-305 — 13 · Advanced networking

1. Network services by function

Service	What it does	Signal in the wording
Azure Bastion	RDP/SSH from the portal (TLS 443) to VMs **without a public IP**; Standard SKU: native client, scaling, file transfer	"manage VMs without public IPs", "no RDP exposed"
NAT Gateway	Outbound Internet access with **fixed public IP(s)**, up to 64k SNAT ports per IP, zonal	"predictable outbound IP", "SNAT port exhaustion"
DDoS Protection	Network Protection (per VNet, all public IPs, ~3k €/month) / IP Protection (per IP); Basic free and always on	"protect public IPs from volumetric attacks", "cost-effective for few IPs" → IP Protection
Azure Firewall	Managed L3-L7 firewall; Standard (FQDN, threat intel) / Premium (TLS inspection, IDPS, URL filtering) / Basic (SMB)	"FQDN filtering", "central inspection", "TLS inspection" → Premium
Firewall Manager	Centralized firewall policies for several firewalls / secured hubs	"manage policies across hubs"
Private Endpoint	NIC with a private IP pointing to a PaaS service; blocks the public endpoint; needs a Private DNS zone privatelink.*	"no public endpoint", "access from on-prem via ER/VPN"
Private Link Service	Expose **your own service** (behind a Standard LB) to consumers in other VNets/tenants via Private Endpoint	"expose our service privately to customers"
Service Endpoint	The subnet reaches the PaaS over the backbone; the service keeps its public IP but accepts only that VNet	"restrict storage to a VNet" without requiring a private IP
DNS Private Resolver	**Hybrid** conditional DNS resolution between on-prem and Azure without DNS VMs (inbound endpoint: on-prem → Azure; outbound + ruleset: Azure → on-prem)	"on-prem must resolve private endpoints", "replace DNS VMs"
Private DNS Zones	Private zones linked to VNets; VM auto-registration	"name resolution across VNets"
Azure DNS	Public zones; alias records	"host public domain"
Route Server	**BGP** exchange between NVAs and the VNet; no manual UDRs; also with ER/VPN gateways	"NVA advertises routes dynamically"
Network Function Manager	Deploys network functions on Azure Stack Edge	Common distractor
Virtual Network Manager	Centralized management of connectivity (mesh/hub-spoke) and admin security rules at scale	"many VNets, connectivity at scale"
Network Watcher	Diagnostics: IP flow verify, NSG diagnostics, next hop, connection troubleshoot, packet capture, flow logs, topology	"allowed or denied", "why can't VM1 reach"
Traffic Analytics	Analytics over flow logs in Log Analytics	"visualize traffic patterns"
Connection Monitor	Latency/loss between endpoints (Azure, on-prem, Internet)	"monitor ExpressRoute latency"

2. ExpressRoute in detail

Concept	Detail
Peering	**Private peering** (VNets) · **Microsoft peering** (Microsoft 365, PaaS with public IP, requires NAT and route filters) · (Public peering retired)
Circuit	Redundant pair of cross-connections by design; SLA 99.95 %; SKUs Local / Standard (same geo) / **Premium** (global, >10 VNets, more routes)
ExpressRoute Direct	10/100 Gbps directly to Microsoft without a provider; supports **MACsec**
Global Reach	Connects two on-prem sites **to each other** through their ER circuits
FastPath	Traffic bypasses the gateway → lower latency and higher throughput (Ultra Performance / ErGw3AZ)
Gateway SKUs	Standard, HighPerformance, UltraPerformance, ErGw1/2/3AZ (zone-redundant)
Encryption	**Not encrypted by default**; IPsec VPN over ER (private peering) or MACsec (ER Direct)
Failover	S2S VPN over the Internet as backup ("no MPLS"); or a second circuit with another provider
ER + VPN coexistence	Same gateway subnet, VPN as backup; BGP prefers ER
Virtual WAN	ER only with Standard SKU

3. VPN Gateway in detail

Concept	Detail
Site-to-Site	IPsec/IKE; route-based (recommended, BGP, multi-site) vs policy-based (Basic, single tunnel)
Point-to-Site	SSTP (Windows only) / IKEv2 / **OpenVPN** (cross-platform, Entra ID auth); auth by **certificate** (public root uploaded to Azure, client cert on devices, revocation by thumbprint), RADIUS or Entra ID
VNet-to-VNet	Tunnel between gateways; alternative: peering (no encryption, cheaper)
Active-active	Two public IPs, two tunnels; HA
Zone-redundant SKUs	VpnGw1AZ…
SKUs	Basic (retired for new deployments, no BGP) → VpnGw1-5; throughput and tunnels grow
Gateway transit	Spoke uses the hub's gateway via peering ("Use remote gateways" / "Allow gateway transit")

Concept	Detail
Local Network Gateway	Represents the on-prem site (public IP + prefixes)
Gateway Subnet	Mandatory, /27 or larger recommended

4. Peering and routing

Rule	Detail
Peering **is not transitive**	A↔B and B↔C does not give A↔C; solutions: direct peering, hub with NVA/Azure Firewall + UDRs, Virtual WAN
Global peering	Between regions; no gateway transit towards Basic LB
UDR	Route table per subnet; next hop: VNet, Internet, Virtual appliance (IP), Virtual network gateway, None
Force traffic through the firewall	UDR with next hop = firewall private IP on the source subnets; 0.0.0.0/0 → firewall to force egress
NVA	VM with **IP forwarding** enabled; single point of failure → HA pair behind an Internal LB (HA ports) or Azure Firewall
BGP	Propagation of ER/VPN routes; "Disable BGP route propagation" on the route table so a spoke does not learn on-prem routes
Subnet sizing	Azure reserves **5 IPs**; /29 minimum (3 usable); Gateway Subnet /27; Bastion /26 (AzureBastionSubnet); Firewall /26 (AzureFirewallSubnet)
Overlap	Forbidden between connected networks (peering, VPN, ER)
Service tags	Groups of service IPs (Storage, Sql, AzureCloud) for NSG/Firewall
ASG	Group NICs for NSG rules by app role

5. Network security: what filters what

Tool	Layer	Criteria	Scope
NSG	3-4	IP, port, protocol, service tags, ASG; stateful	Subnet / NIC; most restrictive rule wins by priority
Azure Firewall	3-7	+ **FQDN**, FQDN tags, threat intel, TLS inspection (Premium)	Central hub
WAF	7	OWASP, bots, geo, rate limit	App Gateway / Front Door
DDoS	3-4	Volumetric	Public IPs
PaaS firewall	—	Allowed IPs / VNets; disable public access	Storage, SQL, Key Vault…
JIT VM access	—	Opens management ports for a limited time (Defender)	VMs

6. Private connectivity to PaaS: Private Endpoint vs Service Endpoint

	Private Endpoint	Service Endpoint
Service IP	**Private** in your VNet	Public (optimized route)
Reachable from on-prem (ER/VPN)	✓	✗
Reachable from peered VNets	✓	✗ (only the configured subnet)
Disable public endpoint	✓ possible	The service firewall restricts to the VNet
DNS	Private DNS zone required	None
Cost	Per endpoint + traffic	Free
Signal	"no public endpoint", "on-premises must access"	"restrict to VNet", minimal cost

7. Decision tree

```
What do I need?
├─ Manage VMs without a public IP ................... Bastion
├─ Outbound Internet with a fixed IP ................ NAT Gateway
├─ Protect public IPs from DDoS (few IPs) ........... DDoS IP Protection (many → Network Protection)
├─ Filter by FQDN / central inspection .............. Azure Firewall (+ UDRs)
├─ Inspect TLS / IDPS ............................... Azure Firewall Premium
├─ PaaS without public endpoint, access from on-prem  Private Endpoint + Private DNS zone
├─ Expose my service privately to other tenants ..... Private Link Service
├─ On-prem resolves Azure private names ............. DNS Private Resolver
├─ NVA advertising routes dynamically ............... Route Server
├─ On-prem sites to each other over ExpressRoute .... Global Reach
├─ Encrypt ExpressRoute ............................. IPsec over ER / MACsec (Direct)
├─ ER failover over the Internet .................... S2S VPN
├─ See why a packet does not arrive ................. Network Watcher
└─ Many VNets, connectivity at scale ................ Virtual Network Manager / Virtual WAN
```

8. Signals → answer

Wording	Answer
"RDP without exposing public IP"	Azure Bastion
"outbound IP must not change"	NAT Gateway

Wording	Answer
"encrypt traffic over ExpressRoute"	IPsec over ER (MACsec if ER Direct)
"connect two on-prem sites via ER"	Global Reach
"bypass the gateway for lower latency"	FastPath
"on-prem DNS must resolve Azure private endpoints"	DNS Private Resolver (inbound endpoint)
"NVA must exchange routes with BGP"	Route Server
"filter by FQDN between VNets"	Azure Firewall
"spoke must not learn on-prem routes"	Disable BGP route propagation
"packets allowed or denied"	IP flow verify / NSG diagnostics
"which route a VM uses"	Next hop
"30 VMs, avoid overlap with 172.16.0.0/16"	192.168.0.0/24
"our service consumed privately by customers"	Private Link Service

9. Mistakes to avoid

1. Treating peering as transitive.
2. Service Endpoint when access from on-prem or no public IP is required.
3. Assuming ExpressRoute is encrypted.
4. NSG to filter by FQDN.
5. A single NVA without HA.
6. Forgetting the 5 reserved IPs and the minimum sizes of special subnets.

Cheatsheet AZ-305 — 14 · Security: Defender, Sentinel, Key Vault, B2C and encryption

1. Microsoft Defender for Cloud

Element	What it is	Signal
CSPM (Foundational, free)	Secure Score, recommendations, inventory, basic compliance	"improve security posture", "secure score"
Defender CSPM (paid)	Attack path analysis, cloud security explorer, governance rules	"attack paths"
Defender for Servers (Plan 1/2)	Threat detection on VMs (Azure, Arc), **JIT VM access**, vulnerability assessment, file integrity (P2)	"temporary RDP/SSH", "threat detection on VMs"
Defender for SQL	Vulnerability assessment + Advanced Threat Protection (injection, anomalies)	"detect SQL injection"
Defender for Storage	Malware scanning, suspicious activity	"scan uploaded blobs for malware"
Defender for Containers	AKS runtime, image scanning in ACR	"scan container images"
Defender for Key Vault / App Service / DNS / Resource Manager	Per-service threat detection	
Regulatory compliance	Dashboard per standard (ISO 27001, PCI DSS, NIST, CIS…); based on Azure Policy initiatives	"assess compliance against PCI DSS"
Workflow automation	Logic App on alert/recommendation	"automatic response"
Continuous export	Alerts/recommendations to Log Analytics / Event Hubs (SIEM)	"send alerts to third-party SIEM" → Event Hubs
Azure Arc	Extends Defender to on-prem / other-cloud servers	"hybrid servers"

- Defender is **not** a secret store nor a firewall: it detects and recommends.
- **JIT VM access** ≠ PIM: JIT opens **ports** temporarily; PIM activates **roles**.

2. Microsoft Sentinel

Element	What it is
What it is	Native **SIEM + SOAR**, on top of a Log Analytics workspace
Data connectors	Entra ID, Microsoft 365, Defender, Azure Activity, firewalls, AWS, Syslog/CEF
Analytics rules	Detections (scheduled KQL, ML, fusion) → incidents
Workbooks	Visualization
Hunting	Proactive search with KQL
Playbooks	Response automation with **Logic Apps** (SOAR)
UEBA	User and entity behavior analytics
Signal	"SIEM", "correlate security events across sources", "automate incident response", "security orchestration"

- Sentinel vs Defender: **Defender** = posture and workload protection; **Sentinel** = correlation, investigation and response at organizational scale. They complement each other (Defender is a Sentinel connector).
- Sentinel vs Log Analytics: Sentinel **adds** detection, incidents and automation on top of the workspace.

3. Advanced Key Vault

Feature	Detail
Soft delete	Mandatory (enabled); retention 7-90 days; recover deleted vault/objects
Purge protection	Prevents permanent deletion during the retention period, **even by administrators**; irreversible once enabled; required for CMK on Storage/SQL/Disk Encryption
RBAC vs access policies	RBAC recommended (Key Vault Administrator / Secrets User / Crypto User…); access policies legacy
Networking	Firewall, service endpoints, **Private Endpoint**; "Allow trusted Microsoft services"
SKU	Standard (software) · **Premium** (HSM-backed, FIPS 140-2 Level 2) · **Managed HSM** (Level 3, single-tenant, own HSM pool) · **Dedicated HSM** (Thales, full control, no PaaS integration)
Rotation	Automatic key rotation by policy; auto-renewable certificates with integrated CA (DigiCert, GlobalSign)
Logging	Diagnostic settings → Log Analytics to audit key usage
BYOK / HYOK	Import your own keys (BYOK); HYOK = key never in the cloud (only Managed HSM / AIP scenarios)
Replication	Automatic to the paired region; read-only during failover
References from App Service / Functions	@Microsoft.KeyVault(SecretUri=...) in app settings with managed identity

- "audit key use + you manage keys + always encrypted at rest" (VM disks) → **Azure Disk Encryption** with Key Vault (or SSE with CMK).
- "FIPS 140-2 Level 2" → Premium; "Level 3" → Managed HSM.

4. Microsoft Entra External ID / B2C

Concept	Detail
Entra B2C	Separate tenant for **customers/consumers**: sign-up, social login (Google, Facebook), **OIDC/SAML federation with third-party IdPs**, user flows and custom policies (Identity Experience Framework)
Entra B2B	Invite **partners** as guests into the corporate tenant; collaboration with Microsoft 365/apps
External ID	Current umbrella name (B2B + external tenants for customers)
B2C issues JWTs	APIs validate them (via APIM validate-jwt with the B2C policy's metadata endpoint)
Signal	"customers", "social login", "third-party OIDC identity provider", "consumer-facing app" → B2C; "partner employees access our SharePoint" → B2B; "no guest accounts" → not B2B either: APIM with JWT validation from the external IdP

5. Encryption and data protection: full map

State / need	Technology	Detail
At rest, Storage	SSE (PMK / **CMK** in Key Vault)	Always on; infrastructure encryption for a double layer
At rest, VM disks	**Azure Disk Encryption** (BitLocker/DM-Crypt, keys in your Key Vault, auditable) · SSE with CMK · Encryption at host · Confidential disk	ADE = "you manage the keys"
At rest, SQL	**TDE** (service-managed or CMK); Policy DINE to enforce	Files and backups
Sensitive columns hidden from DBAs	**Always Encrypted** (deterministic allows equality/join; randomized does not)	"group and join on encrypted columns" → deterministic
Mask by role	**Dynamic Data Masking**	Does not encrypt
In transit	TLS 1.2+, HTTPS only, IPsec, MACsec	"secure transfer required"
In use	**Confidential Computing** (Confidential VMs DCasv5/ECasv5, SGX, enclaves, confidential containers)	"code being processed", "protect from cloud operator"
Application keys	Key Vault secrets + managed identity	Never in code
TLS certificates	Key Vault certificates + App Service / App Gateway integration	Auto-renewal
Data in Microsoft 365 / files	Microsoft Purview Information Protection (sensitivity labels, DLP)	Outside AZ-305 unless mentioned
Regulatory immutability	Immutable Storage (time-based Locked / legal hold)	WORM
Anti-ransomware backups	Soft delete + Resource Guard (MUA)	

6. Compliance and security governance

Need	Solution
Assess against a standard (ISO, PCI, NIST, CIS)	Defender for Cloud **Regulatory compliance** (Policy initiatives)
Enforce secure configurations	Azure Policy (Deny/DINE): HTTPS only, minimum TLS, no public IP, encryption
Trust Center / Microsoft audits	**Service Trust Portal** (SOC, ISO reports)
Data residency	Choose region/geo; Policy "allowed locations"
Separation of duties	RBAC + PIM + Resource Guard + Purge protection
Audit of who did what	Activity Log (control plane) + diagnostic logs (data plane) + Entra audit logs
Long retention of security logs (7 years)	Log Analytics archive / export to immutable Storage
Secrets in pipelines	Key Vault + service connections with workload identity federation

7. Decision tree

```
What is asked?
├─ Posture, secure score, recommendations, JIT ..... Defender for Cloud
├─ Threats on VMs / SQL / Storage / containers ..... Defender for <service>
├─ SIEM, multi-source correlation, automated response .... Sentinel (+ Logic Apps playbooks)
├─ Compliance against a standard ..... Defender Regulatory compliance
├─ Secrets / keys / certificates ..... Key Vault (Premium if HSM L2, Managed HSM if L3)
├─ Nobody can delete the vault or the keys ..... Purge protection + soft delete
├─ Customer identity / social / external IdP ..... Entra B2C
├─ Partners with an account in my tenant ..... Entra B2B
├─ Disk encryption with my keys and auditing ..... Azure Disk Encryption
├─ Data in use ..... Confidential Computing
└─ Column not even the DBA can see ..... Always Encrypted
```

8. Signals → answer

Wording	Answer
"temporary access to management ports"	JIT VM access (Defender for Servers)
"detect SQL injection attempts"	Defender for SQL
"correlate security events from Azure, M365 and on-prem firewalls"	Sentinel

Wording	Answer
"automated response to incidents"	Sentinel playbooks (Logic Apps)
"assess against PCI DSS"	Regulatory compliance
"prevent permanent deletion of keys even by admins"	Purge protection
"FIPS 140-2 Level 2 / Level 3"	Key Vault Premium / Managed HSM
"customers sign in with Google"	Entra B2C
"join tables on encrypted columns"	Always Encrypted deterministic
"encrypt while processing"	Confidential Computing
"send Defender alerts to Splunk"	Continuous export → Event Hubs
"alerts when unauthorized login attempts on VMs"	Defender for Servers / Log Analytics security events + alert

9. Mistakes to avoid

1. Defender as a SIEM → Sentinel.
2. Sentinel as a posture tool → Defender.
3. B2B when the wording says "customers" or "no guest accounts".
4. Key Vault Standard for an HSM requirement.
5. Dynamic Data Masking as encryption.
6. Always Encrypted randomized when joins are required.

Cheatsheet AZ-305 — 15 · Data platforms, analytics and integration

1. Data services map

Service	Category	What for	Signal
Azure Data Factory (ADF)	Integration / ETL-ELT	Low-code pipelines, 90+ connectors, mapping data flows (code-free Spark), Self-hosted IR (on-prem), Azure-SSIS IR	"batch ETL", "low-code", "on-prem sources", "SSIS"
Synapse pipelines	Integration	ADF built into Synapse	Same as ADF inside Synapse
Synapse Analytics	Unified analytics platform	Dedicated SQL pools (MPP warehouse), serverless SQL (query the lake), Spark pools, pipelines, Synapse Link	"data warehouse", "query the lake directly"
Microsoft Fabric	Unified analytics SaaS	Lakehouse, Warehouse, Data Factory, Real-Time, Power BI with OneLake and **Direct Lake**	"single SaaS platform", "Power BI without import"
Azure Databricks	Managed Spark	Data engineering, ML, Delta Lake; Unity Catalog / Credential Passthrough	"Spark", "notebooks", "ML at scale"
Azure Stream Analytics	Real-time streaming	**SQL** queries over streams (Event Hubs, IoT Hub, Blob) with time windows; outputs to SQL, Cosmos, Power BI, Functions…	"real-time", "windowing", "SQL-like over streams", "Power BI dashboard live"
Azure Data Explorer (ADX / Kusto)	Log and time-series analytics	Massive ingestion, interactive **KQL** queries over TB in seconds; foundation of Log Analytics and Sentinel	"telemetry analytics", "time series", "interactive ad-hoc over billions of rows"
Event Hubs / IoT Hub	Ingestion	See below	
Data Lake Storage Gen2	Lake storage	Hierarchical namespace, POSIX ACLs, Hadoop	"lake", "Hadoop", "directories by date"
Cosmos DB	Global NoSQL	See cheatsheet 09	
Azure SQL / Hyperscale	Relational	See cheatsheet 09	
Azure Cache for Redis	Cache	Session state, read cache	"near the app"
Microsoft Purview	Data governance	Catalog, classification, lineage, glossary	"discover and classify sensitive data"
Azure Data Share	Sharing	Snapshots to other organizations	"partner", "restricted access to snapshots"
Azure AI Search	Search	Full-text, vector indexes	"search across documents"

2. Data Factory: pieces that come up

Piece	Detail
Pipeline / activities	Copy, Data Flow, Lookup, ForEach, Execute SSIS, Databricks notebook…
Integration Runtime (IR)	**Azure IR** (cloud, managed) · **Self-hosted IR** (installed on-prem/VM for private sources; formerly Data Management Gateway; HA with multiple nodes) · **Azure-SSIS IR** (runs SSIS packages; SSISDB in Azure SQL DB or MI)
Mapping data flows	Visual transformations executed on managed Spark
Triggers	Schedule · Tumbling window (non-overlapping windows, retries, dependencies) · Event-based (Blob created/deleted via Event Grid) · Manual
Linked services / datasets	Connections and structures
Managed VNet + Private Endpoints	Azure IR inside a managed network to reach private PaaS
CI/CD	Git integration (Azure Repos / GitHub), ARM export
Monitoring	Azure Monitor, alerts on pipeline failure

- ADF **moves and transforms**; it does **not** share data with third parties (Data Share) nor analyze in real time (Stream Analytics).

3. Streaming: who does what

Stage	Service
Device telemetry ingestion (bidirectional, device management, DPS)	**IoT Hub**
App event / log ingestion (unidirectional, high volume, Kafka)	**Event Hubs**
Automatic capture to Storage/ADLS by date	**Event Hubs Capture**
Real-time processing with SQL and windows	**Stream Analytics**
Real-time processing with code / Spark	**Functions** (per event) / **Databricks / Synapse Spark** Structured Streaming
Interactive analytics over ingested data	**Data Explorer**
Live dashboard	**Power BI** (streaming dataset from Stream Analytics)
Store results	SQL, Cosmos DB, ADLS
Edge	**IoT Edge** (modules on the device; Stream Analytics on IoT Edge)

- "IoT devices" + "device provisioning / twins / commands to devices" → **IoT Hub**; only massive inbound telemetry without management → **Event Hubs** is cheaper.
- "hot path" (seconds) → Stream Analytics; "cold path" (batch) → ADF/Synapse; "warm path" (minutes, interactive) → Data Explorer.

4. Warehousing and lakehouse: choosing the engine

Need	Engine
Relational MPP warehouse, few heavy queries	Synapse **dedicated SQL pool** (≤128 concurrent)
Relational warehouse, **high concurrency (>128)**, up to 100 TB	**Azure SQL Hyperscale**
Query lake files without loading (pay per TB scanned)	Synapse **serverless SQL**
Spark transformations, Delta Lake, ML	Databricks / Synapse Spark / Fabric
Analytics over Cosmos DB without ETL	**Synapse Link** (analytical store)
Analytics over Azure SQL without ETL	Synapse Link for SQL
Logs / time series at scale	Data Explorer
Power BI over the lake without import	Fabric **Direct Lake**
Saturated Fabric capacity	Scale the SKU (F4 → F8…), optimize refreshes, workspace governance

5. Purview (data governance)

Feature	What it does
Data Map	Scans sources (Azure, on-prem via self-hosted IR, AWS, Power BI) and builds the map
Data Catalog	Asset search, business glossary, owners
Classification	Detects sensitive data (PII, cards) with built-in and custom rules
Lineage	Source → transformations (ADF, Synapse) → reports
Data Estate Insights	Coverage and sensitivity metrics
Signal	"catalog", "classify sensitive data across the estate", "lineage", "data governance"

- Purview ≠ Azure Policy (governs resources, not data) ≠ Defender (security) ≠ Data Share.

6. Databricks: exam points

Topic	Rule
User identity down to ADLS (per-folder ACLs)	**Credential Passthrough** (Personal/Single User compute) or **Unity Catalog** with external locations
On-prem apps calling the workspace, staff turnover	**Service principal** (not PAT, not managed identity outside Azure)
Secrets	Secret scopes backed by Key Vault
Query performance	Photon (engine), not security
Private networking	VNet injection, Private Link
Tiers	Standard vs Premium (granular RBAC, credential passthrough, audit logs)

7. Decision tree

```
What do I do with the data?
├─ Move / transform in batch, low-code ..... Data Factory (Self-hosted IR if on-prem)
├─ Process a real-time stream with SQL ..... Stream Analytics
├─ Process a stream with code ..... Functions / Spark streaming
├─ Ingest from devices with management ..... IoT Hub
├─ Ingest massive app events ..... Event Hubs
├─ Query logs/telemetry interactively ..... Data Explorer
├─ Relational MPP warehouse ..... Synapse dedicated pool (Hyperscale if >128 concurrent)
├─ Query the lake without loading ..... Synapse serverless
├─ Spark / ML ..... Databricks / Synapse Spark
├─ Analytics over Cosmos DB without ETL ..... Synapse Link
├─ Catalog and classify data ..... Purview
├─ Share with a partner ..... Data Share
└─ Single SaaS platform with Power BI ..... Fabric
```

8. Signals → answer

Wording	Answer
"low-code batch ETL", "mapping data flows"	Data Factory
"run existing SSIS packages"	Azure-SSIS IR
"ADF must read SQL Server on-prem"	Self-hosted Integration Runtime
"real-time aggregation with time windows, SQL syntax"	Stream Analytics
"device twins, provisioning, cloud-to-device"	IoT Hub
"interactive queries over billions of log rows"	Data Explorer
"query JSON files in the lake directly"	Synapse serverless SQL
"50 TB relational + 300 concurrent reads"	Hyperscale
"near real-time analytics on Cosmos DB"	Synapse Link

Wording	Answer
"user identity enforces folder ACLs in Databricks"	Credential Passthrough
"classify sensitive data across sources, lineage"	Purview
"give Fabrikam restricted access to snapshots"	Data Share
"reduce table scans in Table Storage"	Migrate to Cosmos DB with a suitable partition key

9. Mistakes to avoid

1. Stream Analytics for batch ETL → ADF.
2. Event Hubs when device management is required → IoT Hub.
3. Synapse dedicated pool with high concurrency → Hyperscale.
4. Purview to govern resources → Policy.
5. Data Factory to share with third parties → Data Share.
6. PAT in Databricks with staff turnover → service principal.

Cheatsheet AZ-305 — 16 · Additional compute, Azure Arc and Infrastructure as Code

1. Compute services that were missing

Service	What it is	Signal
Azure Virtual Desktop (AVD)	Managed multi-session Windows 10/11 remote desktops and apps; host pools (pooled / personal), application groups (Desktop / RemoteApp), workspaces; **FSLogix** for profiles on Azure Files/NetApp	"remote desktops", "multi-session", "profiles", "publish RemoteApps"
Azure Batch	Large-scale parallel compute with its **own scheduler**; VM pools (incl. Spot), jobs and tasks	"parallel processing", "render", "no third-party scheduler"
Azure CycleCloud	Orchestration of HPC clusters with **third-party schedulers** (Slurm, PBS, LSF)	"HPC", "third-party scheduler"
HPC (H/HB/HC series, InfiniBand)	VMs for MPI / simulation	"low-latency interconnect", "MPI"
Azure Spring Apps	Managed PaaS for **Java Spring Boot** (Enterprise: Tanzu)	"Spring Boot microservices without managing infrastructure"
Static Web Apps	Hosting for static frontends (React, Angular…) with **integrated Functions** as the API, CI/CD from GitHub/Azure DevOps, global CDN, built-in auth	"static site + serverless API", "JAMstack", "deploy from GitHub"
App Service (Linux / containers)	Web apps from a container image	"custom container in App Service"
Azure Functions Durable / Logic Apps	See cheatsheet 06	
Dedicated Hosts	Dedicated physical servers	"no shared hardware"
Azure Stack HCI / Hub / Edge	Azure in your datacenter (HCI: hyperconvergence; Hub: Azure services on-prem; Edge: AI/IoT appliance)	"run Azure services on-premises", "disconnected"
Azure Local	Current name of Azure Stack HCI	

2. Azure Arc

Capability	What it provides
Arc-enabled servers	Windows/Linux servers **on-prem or in other clouds** as Azure resources: RBAC, tags, Policy (guest configuration), **Azure Monitor Agent**, Defender for Servers, Update Manager, Automation, extensions
Arc-enabled Kubernetes	External clusters managed from Azure: GitOps (Flux), Policy, Defender, Monitor
Arc-enabled data services	**Azure SQL Managed Instance** and PostgreSQL **running on your on-prem Kubernetes**, with Azure billing and management; direct or indirect (disconnected) mode
Arc-enabled SQL Server	On-prem SQL Server inventoried and protected (Defender, best practices, Purview)
Arc-enabled VMware / SCVMM	Manage VMware/Hyper-V VMs from Azure
Signal	"on-premises servers", "AWS instances", "single pane", "apply Azure Policy to on-prem", "monitor hybrid with AMA"

- Arc **does not migrate** and **does not replicate**: it manages resources where they are.
- Arc + AMA + central Log Analytics = hybrid monitoring (cheatsheet 07).
- Arc-enabled servers require the **Connected Machine agent** and outbound HTTPS to Azure.

3. Infrastructure as Code

Tool	What it is	Detail
ARM templates	Native declarative JSON	Verbose; the foundation of everything
Bicep	Microsoft's declarative DSL that compiles to ARM	More readable, modules, types; **recommended**; stateless (Azure is the source of truth)
Terraform	Multi-cloud (HashiCorp), HCL, **state file**	Multi-cloud or the team already uses it; state in Storage with locking
Pulumi	IaC with programming languages	Niche
Template Specs	ARM/Bicep templates **stored and versioned as Azure resources**, shareable via RBAC	Replaces Blueprints for template reuse
Deployment Stacks	Group deployed resources as a managed unit; **deny settings** protect against changes outside the stack; cascading delete	Replaces Blueprints for lifecycle governance
Azure Blueprints	Retired (July 2026); definition at MG, assignment per subscription	See cheatsheet 04 if it appears
Deployment modes	**Incremental** (default: adds/updates, does not delete) vs **Complete** (deletes what is not in the template)	"remove resources not in template" → Complete
Deployment scopes	Resource group · Subscription · Management group · Tenant	Policy/RBAC at MG level → deployment at MG scope
What-if	Preview changes	"preview changes before deploying"

Tool	What it is	Detail
Azure Resource Graph	KQL queries over all resources	"query resources across subscriptions"

- **Configuration version history** → templates (Bicep/ARM) in **Git** (Azure Repos / GitHub).
- Configuration **inside** the VM at provisioning → **Custom Script / DSC / Azure Automation State Configuration**, extensionProfile in VMSS; **Azure Image Builder / Compute Gallery** for golden images.
- **Azure App Configuration** + Key Vault references → centralized app configuration and feature flags.

4. DevOps and deployment

Piece	Detail
Azure DevOps	Repos (Git), Pipelines (YAML), Boards, Artifacts, Test Plans
GitHub	Actions (CI/CD), Advanced Security, Codespaces
Service connection / workload identity federation	Pipelines authenticate to Azure **without secrets** (OIDC)
Environments + approvals	Gates between stages
Deployment slots (App Service)	Swap with no downtime; warm-up; rollback
Blue/green and canary	Slots, Traffic Manager weighted, Front Door, Container Apps revisions, Istio
Rolling upgrade (VMSS)	Update in batches with health probe
Feature flags	App Configuration
Self-hosted agents	When the pipeline needs a private network / specific software
Artifacts / Container Registry	Versioned packages and images; ACR **Tasks** for builds; geo-replication (Premium)

5. VM images and configuration

Need	Solution
Reusable, versioned, multi-region golden image	**Azure Compute Gallery** (generalized/specialized images, replicas per region)
Automate image creation	**Azure Image Builder**
Post-boot configuration	Custom Script Extension, DSC, cloud-init (Linux)
Continuous desired state	Azure Automation State Configuration / **Machine Configuration (Policy guest configuration)**
Patching	**Azure Update Manager** (Azure + Arc)
Inventory and changes	Change Tracking and Inventory

6. Decision tree

```
What is asked for?
├─ Multi-session remote desktops / apps ..... Azure Virtual Desktop
├─ Managed parallel compute ................. Batch
├─ HPC with Slurm/PBS ....................... CycleCloud
├─ Spring Boot without managing infra ....... Azure Spring Apps
├─ Static frontend + serverless API from GitHub ..... Static Web Apps
├─ Manage servers/K8s/SQL outside Azure ..... Azure Arc
├─ SQL MI on my on-prem Kubernetes .......... Arc-enabled data services
├─ Define infrastructure with history ....... Bicep in Git
├─ Multi-cloud IaC .......................... Terraform
├─ Reuse templates with RBAC ................ Template Specs
├─ Protect resources from changes outside the deployment .... Deployment Stacks (deny settings)
├─ Delete whatever is not in the template ... Complete mode
├─ Pipeline without secrets ................. Workload identity federation
├─ Zero-downtime deployment on App Service .. Slots
└─ VM image versioned per region ............ Compute Gallery
```

7. Signals → answer

Wording	Answer
"multi-session Windows desktops", "FSLogix profiles"	Azure Virtual Desktop
"render farm, thousands of tasks, no scheduler"	Batch
"Spring Boot", "Tanzu"	Azure Spring Apps
"React app + Azure Functions API, GitHub"	Static Web Apps
"apply Azure Policy to AWS EC2 instances"	Azure Arc-enabled servers
"SQL Managed Instance on-premises Kubernetes"	Arc-enabled data services
"readable, modular, Azure-only IaC"	Bicep
"state file", "multi-cloud"	Terraform
"share templates with teams via RBAC"	Template Specs
"prevent deletion outside the deployment"	Deployment Stacks

Wording	Answer
"delete resources not in template"	Complete mode
"preview changes"	What-if
"query all VMs across 50 subscriptions"	Azure Resource Graph
"golden image, multiple regions"	Compute Gallery
"patch Azure and on-prem servers centrally"	Update Manager (+ Arc)

8. Mistakes to avoid

1. Batch for HPC with a third-party scheduler → CycleCloud.
2. Arc as a migration tool → it only manages.
3. Blueprints as the modern answer for reuse → Template Specs / Deployment Stacks.
4. Incremental when asked to delete what is left over → Complete.
5. ZIP or images as "configuration definition with history" → Bicep in Git.
6. Compute Gallery for App Service → it is for VMs.

Cheatsheet AZ-305 — Azure SQL (PaaS): HA and DR

Applies to **Azure SQL Database** and **Azure SQL Managed Instance**. Azure manages the engine; you choose the mechanism.

1. The four mechanisms in one table

	Zone-redundant HA	Active geo-replication	Auto-failover group	Geo-restore
Protects against	Node / zone (data center) failure	Region outage	Region outage	Region outage (data only)
Reach	Within the region	Across regions (or same)	Across regions	Across regions
Replication	Synchronous	Asynchronous	Asynchronous	Backup copy
RPO	**0**	Seconds	Seconds	Up to 1 hour
RTO	Seconds	Seconds-minutes	Seconds-minutes	**Hours**
Failover	Automatic	**Manual**	**Automatic** (or manual)	Manual (restore)
Connection string	Does not change	**Changes** (different server)	**Does not change** (listener)	Changes (new server)
Scope	1 database	1 database	**Several DBs / whole MI**	1 database
Readable secondaries	Yes on Business Critical / Premium	Up to **4**, readable	**1**, readable (.secondary)	No
Managed Instance	Yes	**No**	**Yes** (the only geo-DR for MI)	Yes
Cost	Included in the tier	Each secondary is paid	The secondary is paid	Backup storage only

2. Zone-redundant HA (local high availability)

- It is not DR: replicas within the **same region**, spread across availability zones.
- **General Purpose zone-redundant:** ZRS storage + relocatable compute. Survives 1 zone.
- **Business Critical / Premium (DTU):** internal Always On AG, **primary + 3 synchronous secondaries**, local SSD. Survives **2 zones** down. Read replica included.
- **Standard (DTU) and Basic: do not** support zone redundancy.
- Signals: "zone outage", "data center failure", "no data loss", "minimize I/O latency".

3. Active geo-replication

- Creates up to 4 **readable secondaries** in other regions, continuous asynchronous replication.
- **You** decide the failover (or your script). When promoting, the app must point to the **new logical server**.
- Only **Azure SQL Database**, one DB per configuration.
- Signals: "readable secondaries", "manual failover", "a single database", "distribute reads across regions".
- Trap: if the wording says "without administrative intervention" → it is **not** this one.

4. Auto-failover group

- Active geo-replication + automation + connection abstraction.
- Groups **several DBs of a logical server** (or a whole MI): they fail over **together**.
- Two fixed **listener endpoints**: .database.windows.net → read/write (current primary) .secondary.database.windows.net → read-only
- **Automatic** failover with configurable grace period, or manual.
- **1 secondary**, mandatorily in **another region**.
- It is the **only** geo-DR mechanism for **Managed Instance**.
- Signals: "automatic failover", "without administrative intervention", "without modifying the app / connection string", "several DBs must fail over together", "Managed Instance".

5. Geo-restore

- Automatic backups (weekly full, diff every 12-24 h, log every 5-10 min) copied to the paired region (RA-GRS).
- On disaster: **create a new DB** by restoring the latest available backup.
- Cheap, but **RTO of hours** and **RPO of up to 1 h**.
- Signals: "minimize costs", "recover the data", tolerates hours of downtime.
- Trap: it is never the answer to "minimize downtime".

6. Firewall and failover groups (exam detail)

- **Database-level** rules (sp_set_database_firewall_rule) → **travel with the replica** on failover.
- **Server-level** rules → are **not** replicated to the secondary server.
- Symptom "one DB connects after the failover and another gets a firewall error" → mix of levels.

7. How they combine (typical mission-critical design)

```
Business Critical zone-redundant  (local HA, RPO 0, automatic)
        +
Auto-failover group to another region  (DR, RPO seconds, automatic)
        +
Automatic backups               (last safety net: corruption / deletion, which replication does NOT protect against)
```

8. Decision tree

```
What must survive?
├─ Node / zone ..................... Zone-redundant (Premium / Business Critical)
├─ Region, manual control, reads ... Active geo-replication
├─ Region, automatic, app untouched  Auto-failover group
├─ Region, data only, minimum cost . Geo-restore
└─ It is Managed Instance .......... Auto-failover group (there is no other)
```

9. Signals → answer

Wording	Answer
"zone outage" + "no data loss"	Zone-redundant / Premium / Business Critical
"readable secondaries in multiple regions"	Active geo-replication
"manual failover"	Active geo-replication
"automatic failover" / "without admin intervention"	Auto-failover group
"connection string must not change" / "minimize changes to the app"	Auto-failover group
"multiple databases fail over together"	Auto-failover group
"Managed Instance" + regional DR	Auto-failover group
"minimize costs" + tolerates hours	Geo-restore
"restore to another region"	Geo-restore

10. Mistakes to avoid

1. Believing that active geo-replication does automatic failover → **no**, that is the failover group.
2. Choosing geo-restore when asked to minimize downtime → RTO of hours.
3. Choosing Standard (DTU) for zone redundancy → **it does not support it**.
4. Choosing active geo-replication for Managed Instance → **it does not exist** on MI.

Cheatsheet AZ-305 — SQL Server on VMs (IaaS): Always On, DR and migration

Applies to **SQL Server installed on virtual machines** (Azure or on-premises). You manage the engine and build HA/DR with the product's technologies. **"Always On"** is the brand of **two** different technologies: **FCI** and **Availability Groups**. They are not the same thing.

1. The technologies in one table

	FCI	Synchronous AG	Asynchronous AG	Distributed AG	Log Shipping	Azure Site Recovery
Protects	Whole instance	Database(s)	Database(s)	A whole AG	Database	Whole VM
Data copies	**1** (shared)	N	N	N	2+	2
Storage	**Shared** (S2D, shared disks, Premium Files)	Own per replica	Own per replica	Own per AG	Own	Own
Replication	—	Synchronous	Asynchronous	Asynchronous	Log backups in batches	Asynchronous (blocks)
RPO	0	**0**	Seconds	Seconds	Minutes	Minutes
RTO	Sec-min	Seconds	Minutes	Seconds	Minutes	Minutes
Failover	Automatic	Automatic	Manual	Manual	Manual	Orchestrated (Recovery Plan)
Readable secondaries	No	Yes	Yes	Yes	Standby only	No
Cross-version	No	No	No	**Yes** (to a higher version)	**Yes**	—
Cross-platform (Win↔Linux)	No	No	No	**Yes**	**Yes**	—
Cluster manager	WSFC	WSFC / Pacemaker	WSFC / Pacemaker	One per AG	No	No
Reach	Local	Local / zones	Regions	Regions / migration	Regions / migration	Regions
Main use	Instance HA	HA with RPO 0	DR	DR and **migration**	Cheap DR and migration	VM DR without touching SQL

2. Always On Failover Cluster Instance (FCI)

- Two or more WSFC nodes; **a single active instance** on **a single shared storage**.
- Protects **everything**: databases, logins, SQL Agent jobs, configuration.
- No replicas → if the storage gets corrupted, there is no copy. Combined with AG for DR.
- In Azure: Storage Spaces Direct, Azure Shared Disks or Premium File Shares as shared storage.
- Signals: "instance-level", "shared storage", "logins and jobs must fail over", "no replicas".

3. Always On Availability Group (AG)

- Each replica has **its own copy** of the data; the primary sends the log to the secondaries.
- Protects **databases** (one or several, as a group). Logins and jobs are **not** replicated (Contained AG in 2022 improves this).
- **Synchronous:** RPO 0, automatic failover, same region/zones. Max. synchronous replicas limited.
- **Asynchronous:** RPO seconds, manual failover, works for another region.
- Up to **8 secondaries**, readable and usable for backups.
- **Listener:** virtual IP/name → in Azure = **Internal Load Balancer + Floating IP + TCP probe** (not HTTP).
- CLUSTER_TYPE = NONE : AG without cluster manager, read replicas only, no automatic failover.
- Signals: "readable secondaries", "no shared storage", "listener", "RPO zero", "several DBs together".

4. Distributed Availability Group

- **AG of AGs:** links two independent AGs (each with its own cluster). Always asynchronous between AGs.
- Supports a **higher version at the target** (2016 → 2019, 2017 → 2022…) and **Windows → Linux**.
- Method documented by Microsoft for **migrations with minimal downtime**: synchronize, manual failover, cutover in seconds.
- Typical target preparation: single-node AG with CLUSTER_TYPE = NONE .
- Signals: "source already has an AG", "migrate to newer version / to Linux", "minimize downtime".

5. Log Shipping

- Periodic log backup on the primary → copy → restore on the secondary (NORECOVERY or STANDBY).

- Simple, cheap, cross-version and cross-platform. Secondary readable only in STANDBY between restores.
- Cutover: last log backup + restore WITH RECOVERY + redirect the app → **minutes**.
- Signals: "no AG at the source", "simple solution", "minutes of downtime acceptable", "low-cost DR".

6. Azure Site Recovery (ASR)

- Replicates the **whole VM** (OS + SQL + everything) to another region; orchestrated failover with Recovery Plans.
- Knows nothing about SQL: RPO of minutes, no readable replicas. Minimum cost (no active VM at the target).
- Signals: "replicate the VM", "relaxed RPO (hours)", "minimize costs", "without touching SQL".

7. Common combinations

Design	Local HA	DR
FCI + asynchronous AG	FCI (instance)	AG to another region
Synchronous AG + asynchronous AG	Synchronous replicas across zones	Remote asynchronous replica
AG + Distributed AG	Local AG	Second AG in another region
Single VM + ASR	None (or Availability Set)	ASR

8. SQL VM rules that show up on the exam

- **Disk host caching:** Log = **None**, Data = **ReadOnly**. If a single policy is required → **None**.
- **OS disk:** Premium SSD **v1** (v2 is not valid as an OS disk).
- **Series with "s"** = Premium storage (DS, Es, Fs…). SR-IOV = Accelerated Networking.
- **SLA:** single VM 99.9 %, Availability Set 99.95 %, Availability Zones 99.99 %.
- **Listener with ILB:** enable **Floating IP**; **TCP** probe on the listener port.

9. Migration tools (summary)

Need	Tool
Assess compatibility beforehand	Data Migration Assistant (DMA)
Migrate to Single DB tolerating downtime	BACPAC
Migrate to MI / VM with minimal downtime (no AG at the source)	Azure Data Studio + SQL Migration extension / DMS online
Migrate 50 DBs to MI offline	Azure Database Migration Service
Migrate from an AG to a higher version or to Linux, minimal downtime	**Distributed AG**
Simple migration, minutes of downtime	Log shipping
Move the whole VM	ASR / Azure Migrate
Oracle/MySQL/DB2 source → SQL	SSMA
Continuous on-prem ↔ Azure connectivity (not migration)	Self-hosted Integration Runtime (Data Management Gateway)

10. Decision tree

```
SQL Server on a VM: what do I need?
├─ Protect the whole instance, with shared storage ..... FCI
├─ Local RPO 0, readable replicas ..... Synchronous AG
├─ DR to another region from an AG ..... Asynchronous AG
├─ Migrate from an AG to another version/platform, minimal downtime  Distributed AG
├─ Cheap DR or simple migration with no AG at the source ..... Log shipping
└─ Replicate the VM without touching SQL, relaxed RPO, minimum cost  Azure Site Recovery
```

11. Signals → answer

Wording	Answer
"shared storage", "entire instance", "logins and jobs"	FCI
"readable secondaries", "no shared storage", "listener"	Availability Group
"failover without data loss" (local)	Synchronous AG
"existing AG" + "newer version / Linux" + "minimize downtime"	Distributed AG
"no AG in source" + "simple" + "minutes of downtime OK"	Log shipping
"RPO 24 hours" + "minimize costs" + "VM"	Azure Site Recovery
"ILB as listener"	Floating IP + TCP probe

12. Mistakes to avoid

1. Confusing FCI (1 copy, instance) with AG (N copies, databases).
2. Ruling out Distributed AG because it is Windows → Linux: **it is supported**.

3. Choosing multi-region AG when the RPO is hours and minimum cost is required → ASR.
4. HTTP probe on the SQL listener → it must be **TCP**.
5. Choosing Session persistence on the listener's ILB → does not apply; what is needed is **Floating IP**.